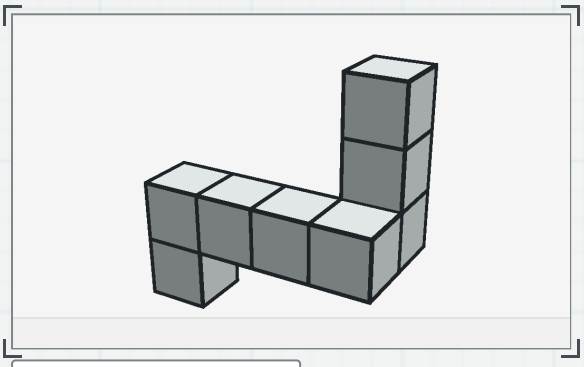
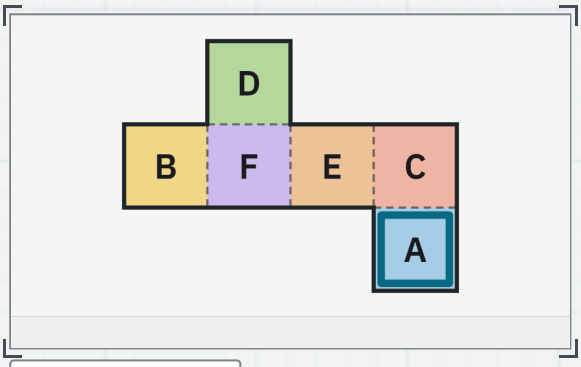
Parallax

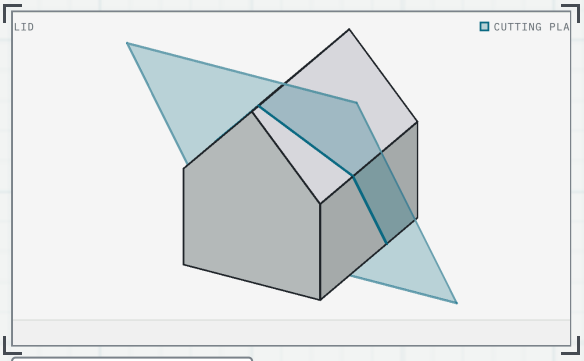
A visuospatial trainer



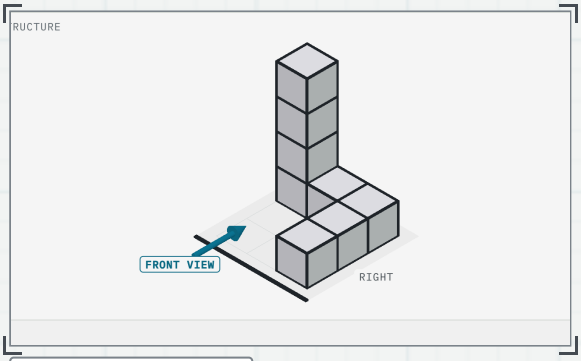
EX-02 · Rotate · Vz SR



EX-05 · Nets · Vz



EX-06 · Slice · Vz



EX-07 · Views · Vz

Built by the Chimera community on Discord

discord.gg/chmr

FREE · OPEN SOURCE · OFFLINE
VERSION 1.1.0
13 NOVEMBER 2026

PARALLAX 1.1.0

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ABOUT THIS GUIDE

This guide explains what Parallax does, how to use it, and what the research behind it shows, including where that research is thin. Its claims come from the project's research notes: four for version 1.0 (evidence, tasks, progression, motivation) and six for 1.1 (path integration, knots, the formats of an admissions test, test overlap, timing, phones). Points those notes could not confirm against a primary source are left out or marked "reported".

Pictures of exercises come from the development harness at mid-level settings; in the app the same views appear inside the player. Most pictures of the app's screens show a record built by about six weeks of simulated practice, played through the real engine: their numbers belong to no real person. The phone pictures and the new screens of 1.1 start from a fresh record.

READING THE NUMBERS

g and **d** are standardised effect sizes: the difference between two groups, or between before and after, in standard deviations. An effect of 0.5 means the average trained person ended half a standard deviation above the average comparison person. **r** is a correlation, from −1 to 1; 0 means none. **N** is the number of people, **k** the number of effects pooled.

Sources appear as (author, year); chapter 13 lists them with links.

Guide text © 2026 Parallax contributors, licensed under CC BY-SA 4.0. Parallax code: MIT licence. Typefaces: IBM Plex Sans and Mono, SIL Open Font License 1.1. Source: the Parallax repository; questions and ideas: discord.gg/chmr.

VERSION 1.1.0 · 13 NOVEMBER 2026

What's new in 1.1

One honest clock, phones as first-class devices, path integration without landmarks, knots, practice in the question formats of an admissions test, and a careful look at which exercises overlap with published tests. A record from 1.0 carries over, with nothing lost.

One clock that counts honestly

Version 1.0 timed a session with four clocks that disagreed. Now one active-time clock runs during items, feedback, show-me and tutorials, and stops for the pause sheet, any open sheet or dialog, a hidden tab, and after 60 seconds without input. Its minutes end Train, Path and Speed sessions, and the HUD, the summary, the Log, the share card and the weekly goal show the same number. Other fixes: next page. Page 27

Phones, first-class

Every scored item of all nineteen exercises and six probes fits a 360 × 640 screen without scrolling; held sideways, items switch to two columns. During an item the header steps aside and the goal and tips wait behind Help. Turning the phone while an item plays voids that trial. Homing was rebuilt for touch, with slower, smoother motion. Page 9

Reckon EX-12

Dead reckoning with nothing to steer by: watch a walk across an empty plain, then point home or walk home. Feedback gives the angle and distance errors, the drift per metre and your habit over the last 20 routes. A community member pointed out that Homing's landmarks correct the very errors this training should work on; Homing stays as the landmark-aided stage before Reckon. Experimental. Page 18

Knots EX-13

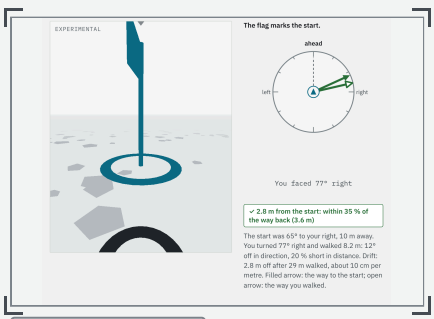
Pull both ends: does a knot remain? Which marked crossing, switched, unties the rope, or ties it? Are two loops linked? Every answer is proved before you see the item, by the moves that take a rope apart, a determinant or the Jones polynomial; Show me replays the moves, checking each one. Page 20

Six DAT-style formats EX-14 to EX-19

Openings, Missing View, Angles, Hole Punch, Painted Cubes and Form practise the six question formats of the Perceptual Ability Test in the Dental Admission Test (DAT®). They join Train only under the DAT-style focus, and Section practice rehearses them in the test's order against one clock, with a review and no score prediction. Page 21

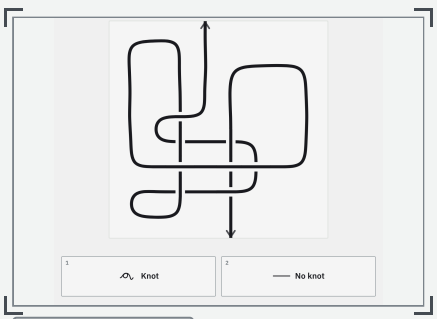
Test hygiene

Practising a kind of puzzle raises scores on tests that use it. Exercises whose formats published tests share carry an "Also in tests" note; About has a Test hygiene section; later Spatial Checks ask one question about outside practice; and a "Taking a test soon?" helper can leave the matching exercises out of Train until your test. Page 40



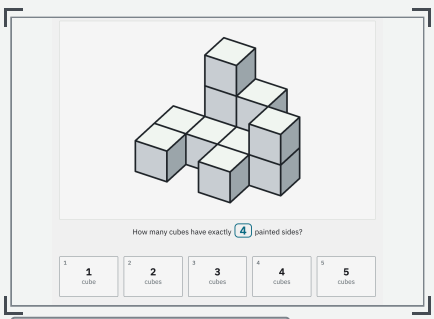
EX-12 · Reckon

After a walk back: the errors in direction and distance, and the drift.



EX-13 · Knots

Pull both ends: does a knot remain?



EX-18 · Painted Cubes

How many cubes have exactly 4 painted sides?

DAT® and Dental Admission Test are marks of the American Dental Association. Parallax is not affiliated with or endorsed by the ADA. Every item is generated by the app; none is taken from the DAT.

Fixed in 1.1

An audit of version 1.0 played real sessions in a browser, with pauses, hidden tabs and idle stretches, and found these problems. Each is fixed, and a test now plays five-minute sessions with such stretches and checks that every screen shows the same minutes.

IN 1.0	IN 1.1
The countdown ("about 5 min left") stood still while you answered quickly, and sessions could end with minutes "left".	The HUD counts down active time, refreshed every few seconds, and a session ends within about one item of its length.
Pauses, hidden tabs and idle time ran the session down, while the pause sheet said the clock had stopped.	The clock stops for the pause sheet, any sheet or dialog, a hidden tab and after 60 seconds without input, so the pause sheet tells the truth.
Practice minutes dropped any gap of more than 90 seconds whole, and counted idle gaps just under it.	Practice minutes are the clock's reading.
One session showed four different minute counts ("0.7 min", "1 min").	One format everywhere: "under 1 min", "7 min", "1 h 05 min". Progress shows your practice totals.
A reload split one sitting into two sessions; reopening the app could start a session by itself.	A reload never starts a session; it shows the last summary, or Today.
The Daily Challenge: a reload or "End now" spent the day's one scored attempt; unanswered items showed as wrong; Today mixed the UTC and the local day.	The attempt is spent only when all 8 items are answered; a partial attempt resumes; unanswered items show as "–"; one local day everywhere, and Today shows the challenge's number.
Speed sessions were sized by item count, not by their deadlines.	Speed plans from its deadlines and ends on time.
Progress counted skipped and voided items as answers; a Spatial Check counted as a practice session for the weekly goal.	Only answered items count; a check never counts as practice.
"Measured" was awarded for a Threshold run stopped after 3 items.	Threshold and Speed achievements need a complete run.
Two "Export trials" buttons wrote two different files under one name.	One CSV format, with the screen class, the input and a second value added.
The Sounds setting did nothing.	Short synthesized tones after an answer, off by default, never while an item plays.
Smaller slips: Today too wide at 150 % text on a phone; Today numbering a session's exercises in an order the session did not follow; the item counter one ahead during feedback; the deadline bar jumping back to full after an answer, and a deadline timed from a few milliseconds after the item appeared; Pause not saying Resume; Threshold results without units; Path asking twice before its first item; the Spatial Check's time estimate.	All fixed. A check now says honestly that it takes 15 to 25 minutes.

YOUR RECORD

Version 1.1 stores the record in format 2. A 1.0 record upgrades when 1.1 first opens it, with nothing lost; a 1.0 tab still open elsewhere then runs from memory instead of overwriting the newer record. Every trial gains the screen class (small or large) and the input (mouse, touch, pen or keys), so that results can be compared like with like, and room for a second number where an exercise reports one. Homing's items changed, so its new trials are logged as version 2.

ALSO NEW

- The Daily Challenge follows your local date and leaves out Track, Homing, Reckon and the six test formats, so it is the same on every device.
- 21 achievements: one more, for a full Section practice.
- Checks and sessions record the screen they were taken on, and Progress says when a check was taken on another kind of screen than your starting checks.

CHAPTER 01

What Parallax is, and what it is not

Parallax is a free, open-source web app for practising visuospatial skills: picturing objects as they turn, fold, are cut, or are seen from somewhere else. It works offline in your browser, on a computer or a phone, and everything you do stays on your device.

WHAT PRACTICE CAN AND CANNOT DO

- **Practice improves the practised tasks, and often similar spatial tests.** Across 206 training studies, spatial practice raised spatial test scores by about half a standard deviation (g = 0.47), and spatial tests that were never trained gained about as much as the trained ones, for at least several weeks (Uttal et al., 2013). Part of any raw gain is familiarity: untrained control groups improved too.
- **Effects on grades, work, sport or IQ are not established.** Gains in school mathematics are small (g = 0.28) and were shown mainly in children and teenagers (Hawes et al., 2022). In the one randomised trial with college students, training raised the physics grade a little (d = 0.32); eight months later there were no differences on spatial tests, later grades or chosen majors (Miller & Halpern, 2013). There is no good evidence that spatial training raises intelligence.
- **Parallax measures your transfer instead of promising it.** The Spatial Check uses tasks you never practise, starts from two baselines, and compares your change with a non-spatial control task and with what familiarity alone would give. When those cannot be told apart, it says "unclear yet", which is not the same as no progress.

What you get

- **Nineteen exercises:** rotation in 3D and 2D, paper folding, cube nets, cross-sections, views, perspective taking, spatial span, hidden figures, knots, navigation, tracking as attention practice, and six opt-in formats of an admissions test.
- **A new seed every time,** so each item is new and checked by a solver: exactly one right answer, wrong options that differ visibly, nothing decisive hidden.
- **Five ways to practise and one to measure, on one record:** Train, Path, Threshold, Speed, Free play and the Spatial Check; plus a community Daily Challenge, and Section practice for the six test formats.
- **Feedback that explains.** After a miss, a slow "show me" replays the solution, drawn from the same model that graded your answer.
- **Honest progress:** levels from 1 to 100 with a likely range and the real settings you handle, a weekly goal with grace, achievements for skills.
- **Short tutorials** (Watch, Try, Go) and two strategy cards per exercise, with advice on when to switch.

Who it is for

Adults and teenagers who enjoy spatial puzzles, and anyone curious to measure their own change honestly.

What it is not

- **Not a test of intelligence, and not a medical or diagnostic tool.** It shows no IQ, no index, no percentile and no "brain age".
- **Not a copy of a published test.** No exercise clones Wechsler Block Design or Visual Puzzles, and no item comes from a published test.
- **Not a ranking.** Levels compare you with your own past on one exercise. There are no leaderboards and no points for time spent.
- **Not all one thing.** Track is labelled attention practice: following moving objects is linked to attention and working memory rather than to visuospatial ability. Angles is perceptual-judgement practice; Homing and Reckon are experimental.
- **Not online:** no account, ads or analytics.

Why the careful wording

Reviews of cognitive training find extensive evidence that trained tasks improve, less for closely related tasks, and little for distant tasks or everyday life (Simons et al., 2016). US consumer-protection guidance judges claims by their overall impression, implied claims included; "may" or "helps" does not rescue an unsupported claim (FTC, 2022). Parallax tries to meet that standard.

CHAPTER 02

Quick start

Five steps for a first week. Together they give you a starting point you can trust and a routine you can keep.

01 Open Parallax

There are three ways, and they work the same:

- **One file.** Download `Parallax.html` and open it in any modern browser. No installation, no internet connection needed.
- **The repository.** Download or clone it and open `index.html`.
- **A web address.** Anyone can host a copy on GitHub Pages (chapter 11 shows how).

Phones work, upright or sideways (page 9). The 3D exercises need WebGL; where it is missing they say so, and the other exercises still work.

02 Meet the Chimera pop-up

Every time Parallax opens, it says that it was built by the Chimera community on Discord, with a button to join and the address, discord.gg/chmr, ready to copy (some previews block new tabs, so the address is always there as text). Come and say hi, share results, suggest exercises. **Start training** closes it.

03 Watch, Try, Go

On your first visit Parallax offers a short tour of three exercises: Rotate, Fold and Track. Every exercise starts the same way the first time you meet it:

- **Watch** a slow demo of one item, ending with its solution.
- **Try** three to five easy practice items. They are not logged, and tips appear when you need them.
- **Go:** an easy scored start, after which the engine takes over.

You can skip at any step. **How to play** replays the tutorial later. In a large study of game tutorials, just-in-time tips beat an up-front manual in the most complex of three games; forced steps added nothing (Andersen et al., 2012). So Parallax keeps text short and never blocks you.

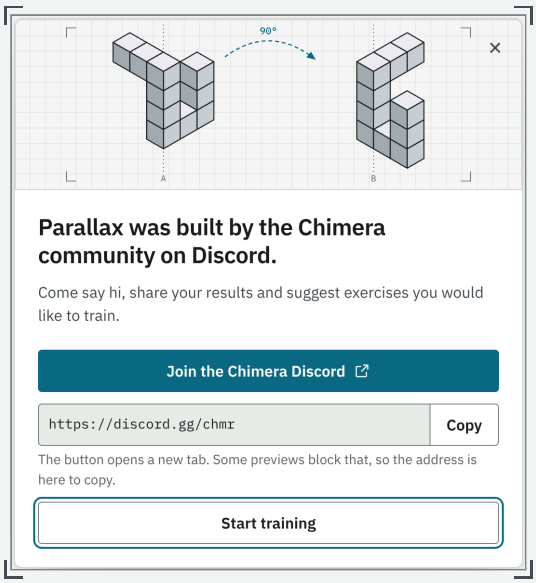


FIG. 2.1 The welcome pop-up, shown on every open.

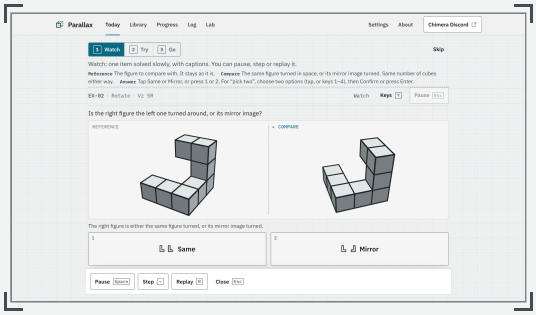


FIG. 2.2 Watch, Try, Go: the first look at an exercise.

04 Take two baseline Spatial Checks

The Spatial Check measures where you start, on tasks you never practise: 15 to 25 minutes, no feedback. Take the first after the tour and the second at your next visit, at least about 12 hours later.

Why two? Retaking any test raises scores on its own, most of all at the second sitting (Scharfen et al., 2018). A second baseline takes up much of that familiarity gain, so later changes mean more. If the two differ a lot, Parallax asks for a third. Skipping only means your transfer report starts later.

05 Train about 20 minutes, three times a week

Today proposes three exercises: a *focus* (one you pin, one not tried yet, or the one where you improve fastest), a *review* (one due for another look) and a *variety* pick (the one practised least).

Twenty minutes is the default (10, 15 and 30 also work); the weekly goal is three sessions (2 to 6), on any days. Several programmes that worked in studies ran about 10 to 20 hours in all, such as 12 hours over six weeks (Miller & Halpern, 2013); more hours have not been shown to help more. Three 20-minute sessions a week make 10 hours in about ten weeks. In older adults, more than three sessions a week brought no extra benefit (Lampit et al., 2014).

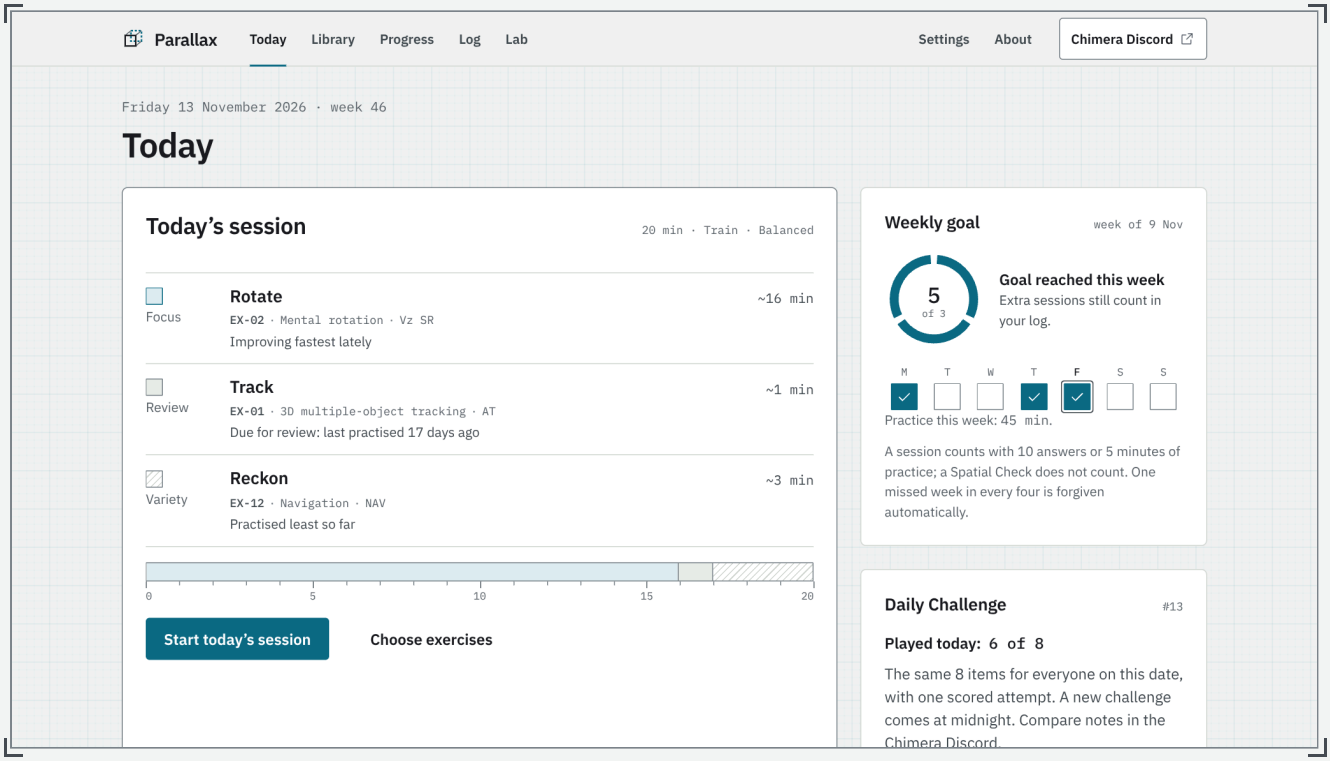


FIG. 2.3 Today, after six weeks of simulated practice: the session plan with its focus, review and variety exercises, the weekly goal and the Daily Challenge.

A first week might look like this

DAY 1 · ABOUT 25 MIN
Open Parallax, take the tour, then the first baseline check.

DAY 2 · 15 TO 25 MIN
The second baseline; a short session if you like.

DAY 4 · 20 MIN
A Train session from Today.

DAY 6 · 20 MIN
A session, then the Daily Challenge; compare notes in the Discord.

A session, step by step

A Train session plays items from its three exercises, aimed at your target. These pictures follow one item, missed on purpose, from the answer to the end of the session.

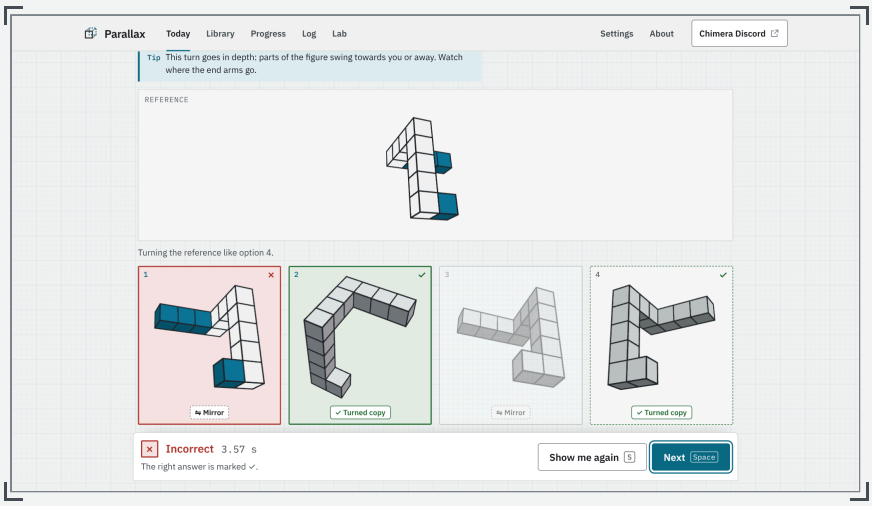


FIG. 2.4 After a wrong answer.

During an item

- **Answer** by tap, click or the keys 1–9; Enter confirms a choice of two; Space goes on; **?** lists the keys.
- **Esc pauses:** the clock stops and the item hides. Pausing while a moving display plays voids that item; it is replaced and never counts.
- **After an answer:** a tick or a cross with a word and a colour (never colour alone), and the right answer. After a miss, **Show me** (S) plays a slow solution you can pause, step and replay; after three automatic replays it waits for you to ask (**Why?**). A right but slow answer offers **See solution**.
- **Skip this item** counts nowhere. Trials where the screen stuttered are void and never count against you.

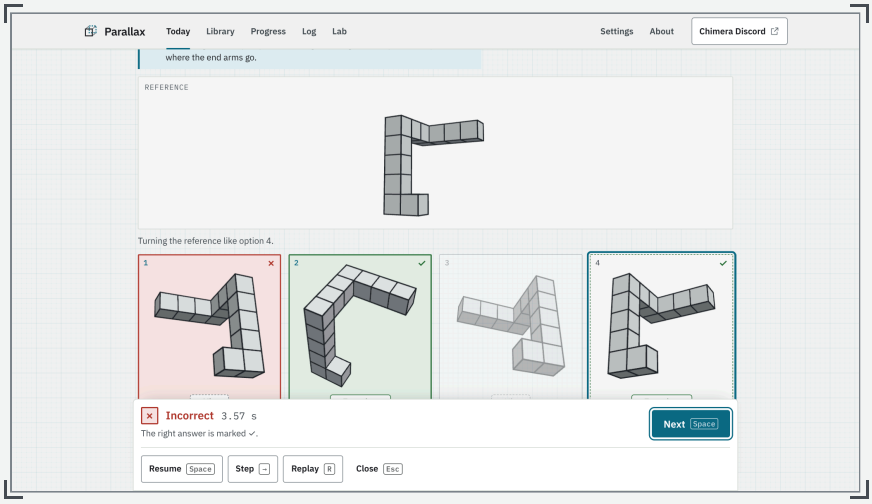


FIG. 2.5 Show me, paused mid-turn.

Show me

A slow replay of the solution, built from the same model that graded your answer: here the reference turns until it matches. Space pauses, → steps, R replays, Esc closes. With reduced motion it steps through keyframes instead of moving smoothly.

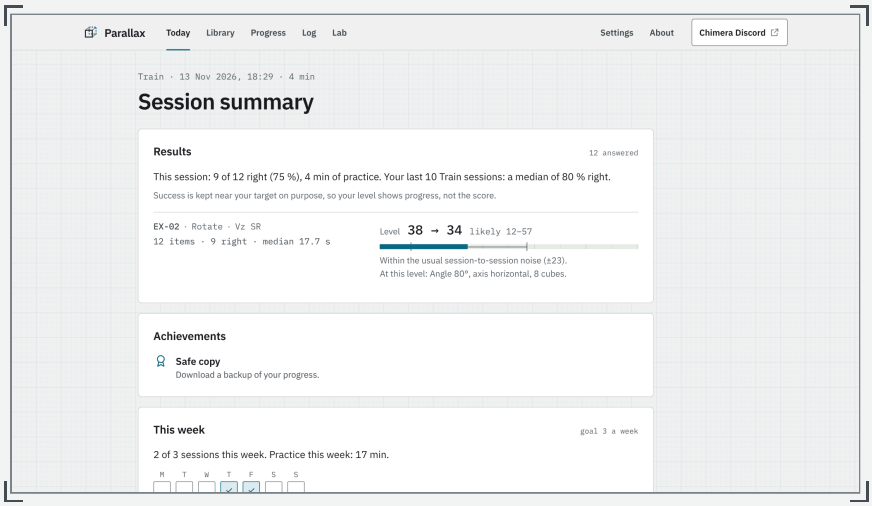


FIG. 2.6 The session summary.

The summary

Results by exercise, compared only with your own past, level changes with the real settings, achievements earned, the week so far, and one suggested next step. Success is held near your target on purpose, so the level, not the score, shows progress. Share makes a card with the Discord link.

On a phone

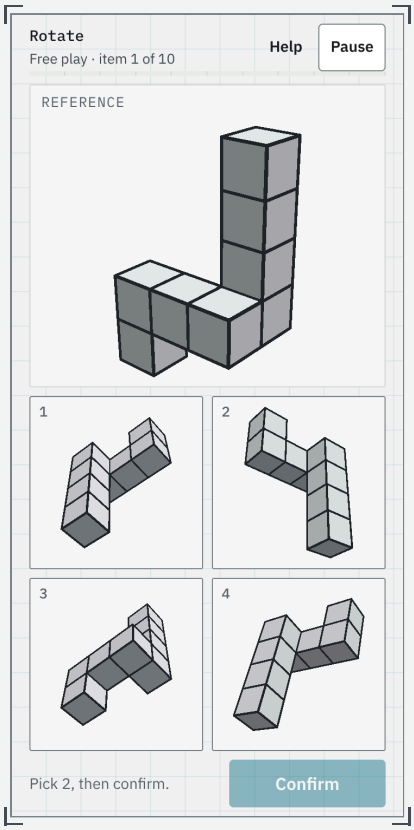


FIG. 2.7 A Rotate item on a phone 360 pixels wide: stimulus, options and Confirm on one screen.

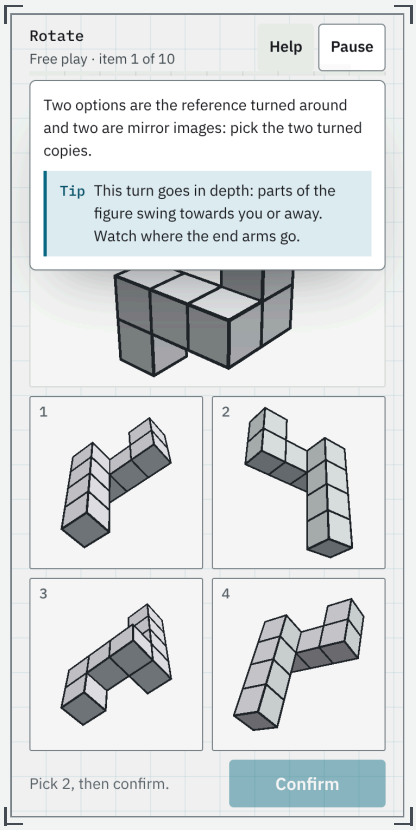


FIG. 2.8 Help shows the goal and the current tip over the item.

Every screen works from 360 pixels wide up, with no sideways scrolling and touch targets of at least 44 pixels; on phones the main screens move to a tab bar at the bottom.

During an item

- The header steps aside; one strip shows the exercise, the count or time left, Help and Pause.
- Every scored item fits a 360 × 640 screen without scrolling: the stimulus and every answer control together.
- Help shows the goal and the tip; the next item closes it.
- Turning the phone while an item plays voids that trial: it is replaced and never counts.

Held sideways

Items switch to two columns, the item on the left and the answers on the right; Track, Span, Orient, Pattern Memory, Homing and Reckon have layouts of their own. Each exercise's items were also checked held sideways, at 844 × 390 pixels.

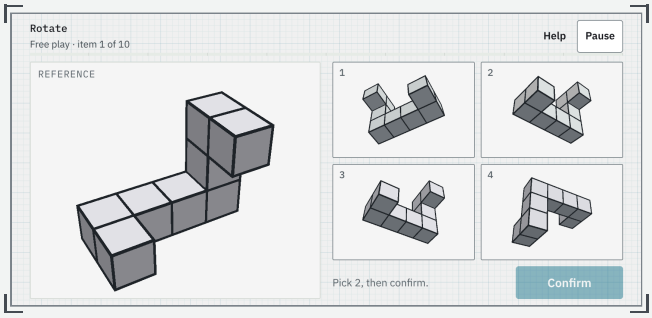


FIG. 2.9 Held sideways, 844 × 390: the item left, the answers right.

Screens and ratings

Track, Homing, Reckon and Angles keep separate ratings for small screens (a short side under 600 pixels) and larger ones, because viewing size changes how hard they are. Every trial, session and check records its screen class and input, so results can be compared like with like.

Where your progress lives

On iPhone and iPad, Safari can clear a site's storage after seven days of use without a visit, and an app's built-in browser, Discord's included, keeps a copy of its own. On these devices the welcome says so before you start, and later offers a backup whenever none was made for 14 days. Adding Parallax to the home screen (Share, then Add to Home Screen) gives it a lasting home; served from GitHub Pages, it then has its own icon. Parallax also asks the browser to keep its storage, which the browser may decline. A backup is the safe choice on every device (chapter 9).

Sharing, settings, and where things are

A share card carries your results and the Chimera link, and no ranking. Where the device allows it, Share… hands the card to another app; Copy text is always there. The Daily Challenge has a one-line share of its own, and friends can swap challenge codes to play exactly the same items (chapter 4).

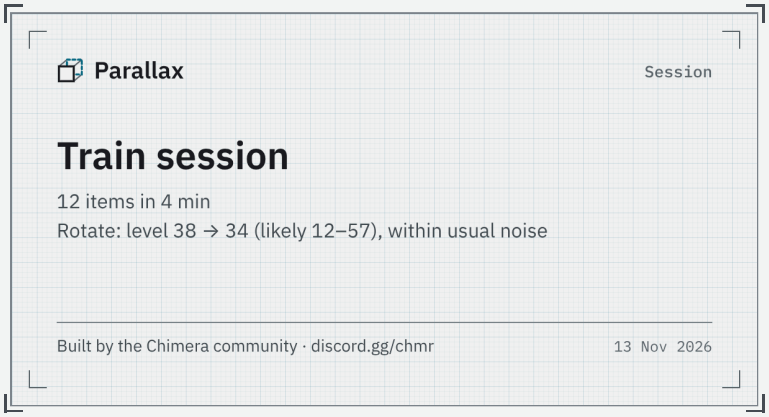


FIG. 2.10 A share card: results and the Chimera link, no ranking.

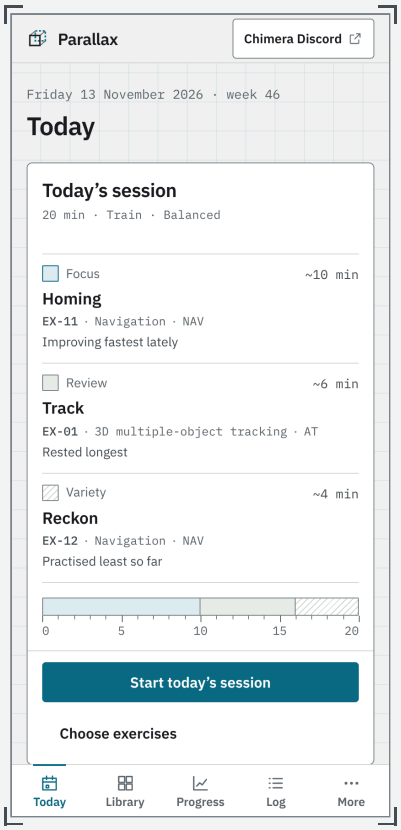


FIG. 2.11 Today on a phone, 412 pixels wide, after six weeks of simulated practice.

SETTINGS WORTH KNOWING

- **Theme:** auto, light or dark. **Text size:** 90 to 150 %.
- **Motion:** follow the device, reduce, or full. Reduce shortens animations around the exercises; moving stimuli, such as the spheres in Track, still move.
- **Sounds** (off by default): a short tone after each answer, high when right and low when not, never while an item plays. **Key hints** on buttons.
- **Session length** and **weekly goal**, as above.
- **Test practice and test hygiene:** the focus on six test formats (page 21) and "Taking a test soon?" (page 40).
- **Your data:** back up, export, import and undo (chapter 9).

Everything is playable by keyboard, touch targets are at least 44 pixels, nothing flashes faster than three times a second, and the device's reduced-motion setting is respected outside the stimulus itself.

WHERE THINGS ARE

- **Today:** the session plan, the weekly goal, the Daily Challenge, the Spatial Check.
- **Library:** every exercise, with How to play, strategies and ways to practise; Section practice.
- **Progress:** levels, curves, personal bests, the Spatial Check report, achievements.
- **Log:** every session and trial; export and import.
- **Lab:** protocols, the research export, the self-experiment.
- **Settings** and **About**, with its Test hygiene section; the Chimera Discord link sits in the header and footer.

CHAPTER 03

The exercises

Ability codes, in plain words

CODE	NAME	WHAT IT MEANS
Vz	Visualisation	Picturing how a complex shape would look after it is turned, folded, cut or partly hidden.
SR	Speeded rotation	Turning simple shapes in your head quickly; the fast cousin of Vz.
CF	Flexibility of closure	Finding a known shape hidden in a busy, distracting pattern.
MV	Visual memory	Holding a picture or a pattern in mind over a short time (under about 30 seconds).
LE	Length estimation	Judging lengths and sizes by eye, without tools; here, how wide angles open (Angles only).
PT	Perspective taking	Imagining what you would see, or where things would be, from another place.
WM	Working memory	Holding and updating information while you use it; here, a sequence of places.
AT	Attentional tracking	Following several moving objects at once.
NAV	Navigation	Keeping track of where you are, and where you came from, as you move.
MA	Mental animation	Working out how the parts of a machine move (a Spatial Check probe only).
VB	Verbal reasoning	Word analogies, used as the Spatial Check's non-spatial control.

Vz, SR, CF, MV and LE are narrow abilities of visual processing in the Cattell–Horn–Carroll model (Schneider & McGrew, 2018; McGrew, 2017); LE is the nearest of them for judging angles by eye. The others are Parallax's own labels. Multiple-object tracking relates to attention and working memory instead (Meyerhoff et al., 2017; Treviño et al., 2021).

Four kinds of spatial task

Researchers sort spatial tasks along two lines (Uttal et al., 2013; Newcombe & Shipley, 2015): **intrinsic** tasks concern one object's shape, **extrinsic** tasks the relations between objects; **static** tasks transform nothing, **dynamic** tasks move or change something. Parallax uses this 2 × 2 as a checklist: the exercises cover three cells, and two Spatial Check probes the fourth (extrinsic-static), which is never trained. It is not a measuring model: in children, a four-factor version did not fit (Mix et al., 2018).

EXERCISE	CODES	KIND OF TASK	FORMATS
EX-01 Track	AT	outside the 2 × 2	still view; slowly orbiting view
EX-02 Rotate	Vz SR	intrinsic-dynamic	same or mirror; pick the two that match
EX-03 Spin	SR	intrinsic-dynamic	same or mirror; pick the two that match
EX-04 Fold	Vz	intrinsic-dynamic	punch, then unfold
EX-05 Nets	Vz	intrinsic-dynamic	opposite face; do the edges meet; does it fold into a cube
EX-06 Slice	Vz	intrinsic-dynamic	name the cross-section
EX-07 Views	Vz	intrinsic-static	structure to view; three views to structure; count the cubes
EX-08 Orient	PT	extrinsic-dynamic	point on an arrow circle
EX-09 Span	MV WM	outside the 2 × 2	forward; backward; a board that turns
EX-10 Hidden	CF	intrinsic-static	which of five shapes is hidden
EX-11 Homing	NAV	extrinsic-dynamic	face the start after a walk, with landmarks on easier levels

Training items are untimed; Speed mode adds deadlines later. Fold, Nets and Views offer off-screen challenges with paper or blocks, for variety. In studies of children, hands-on training went with larger maths gains (Hawes et al., 2022; Gilligan-Lee et al., 2023); these challenges are untested, and Parallax makes no maths claim.

Eight more in version 1.1

EXERCISE	CODES	KIND OF TASK	FORMATS
EX-12 Reckon	NAV	extrinsic-dynamic	point home; walk home; nothing to steer by
EX-13 Knots	Vz	intrinsic-dynamic	does a knot remain; which crossing unties or ties it; are two loops linked
EX-14 Openings	Vz	intrinsic-dynamic	which opening a solid passes through
EX-15 Missing View	Vz	intrinsic-static	two views given: pick the third
EX-16 Angles	LE	outside the 2 × 2	rank four angles by size
EX-17 Hole Punch	Vz	intrinsic-dynamic	fold a small sheet, punch once, unfold
EX-18 Painted Cubes	Vz	intrinsic-static	how many cubes have so many painted sides
EX-19 Form	Vz	intrinsic-dynamic	which solid a flat pattern folds into

EX-14 to EX-19 follow the question formats of an admissions test. They are always in the Library, Free play and Section practice, and join Train only when you switch on their focus (page 21). Together the exercises still cover three cells of the 2 × 2; extrinsic-static tasks remain untrained, for the Spatial Check.

The Library

Every exercise has a Library card: its level with the likely range and the real settings you handle, its Path stage, How to play, the strategy cards, a "where it is used" note, any off-screen challenge, and buttons to practise it in each mode.

Where spatial thinking is used

Each card says honestly where its skill shows up, with the limitation next to it. Rotate's reads: *"Chemists ask the same question about molecules (is this drawing the same molecule or its mirror image?). Engineers turn objects in their heads too. Practice here improves this kind of task and often similar ones; it has not been shown to improve grades or job performance."*

Also in tests

Exercises whose format parts of published tests share, such as Fold, Hidden, Views (counting), Nets, Span, Knots and the six test formats, carry an "Also in tests" note on their page and in How to play (page 40).

Same model, same answer

A solver checks every item before you see it; cheap shortcuts (counting cubes or holes, the odd one out, the shape as seen from here) must score near chance; show-me is drawn from the model that grades your answer. Chapter 8 gives the limits.

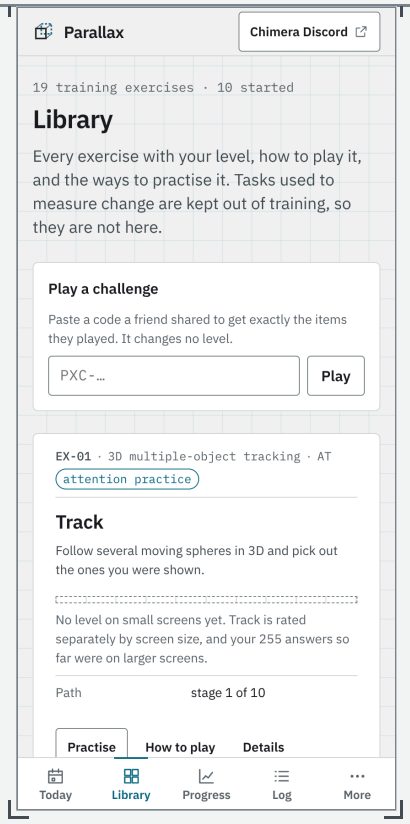


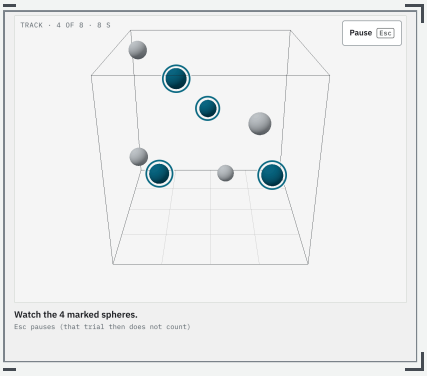
FIG. 3.1 The Library on a phone. Track is rated per screen size, so it has no level here yet.

EX-01 · Track · AT

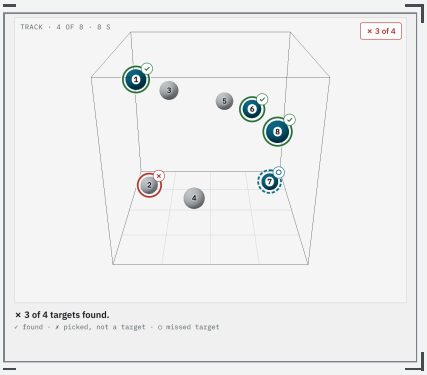
Track

3D multiple-object tracking

attention practice



QUESTION 4 targets of 8 spheres are ringed for 2 s; then all of them move.



FEEDBACK ✓ found, × picked but not a target, ○ a missed target.

TRAINS Following several moving objects at once (AT). Labelled **attention practice**, not visuospatial training: tracking relates to attention and working memory. Ratings are kept apart for small and large screens, because viewing distance changes the difficulty.

HOW TO PLAY Spheres sit in a cube. Some are coloured and ringed for 2 seconds: the targets. Then the marks go and every sphere moves at the same speed, bouncing off the walls and each other. When they stop, each sphere gets a number: pick exactly the targets (tap them or press their numbers), then confirm. A trial is right only when every target is found.

DIFFICULTY GROWS WITH

Speed	0.15 to 2.0 cube widths a second, in 46 steps, each about 6 % faster
Targets	2, 3, 4 or 5
Spheres	6, 8 or 10
Tracking	5, 8 or 12 seconds; the view still, or orbiting slowly

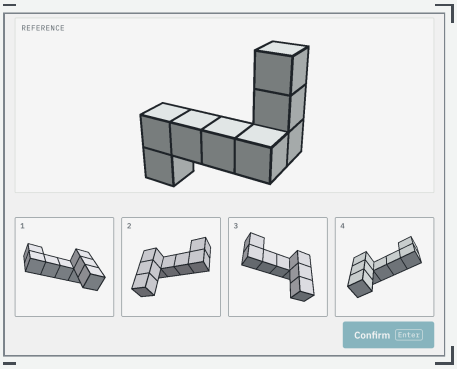
STRATEGIES **One shape:** see the targets as the corners of one shape and watch it stretch, turn and slide, eyes resting near its middle; good for 2 to 4 spread-out targets. **Watch the close passes:** when a sphere comes near a target, watch both until they part; good for fast or crowded displays. SWITCH when the shape falls apart, or when close passes come so often that you lose the picture. When most spheres are targets, follow the others instead.

EVIDENCE Gains on the trained task are large (d = 1.6–2.0), with no transfer to a different tracking task beyond retest (Harris et al., 2020). Studies of a commercial 3D tracking trainer gave very weak evidence for gains in attention or working memory; two of three tests of transfer to sport or driving found nothing, and none were preregistered (Vater et al., 2021). A 2025 soccer study found no transfer to game performance (Romeas et al., 2025).

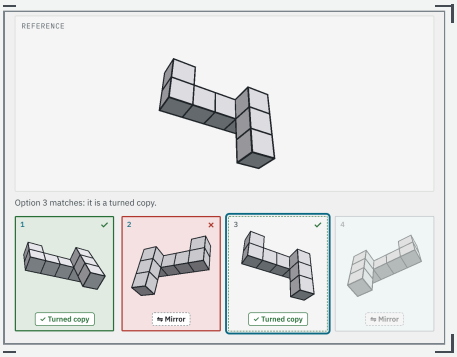
EX-02 · Rotate · Vz SR

Rotate

Mental rotation of 3D block figures



QUESTION Pick two · a 100° turn about a horizontal axis · 8 cubes.



SHOW ME, PAUSED The reference turns into option 3; tags name every option.

TRAINS Visualisation (Vz) and speeded rotation (SR): turning a 3D figure in your head. Intrinsic-dynamic.

HOW TO PLAY Each figure is cubes in four straight arms. **Same or mirror:** is the right figure the left one turned, or its mirror image? Answer Same (1) or Mirror (2). **Pick two:** of four options, two are the reference turned and two are mirror images turned; pick the two copies and confirm. A mirror image has exactly the same cubes, so counting never helps.

DIFFICULTY GROWS WITH

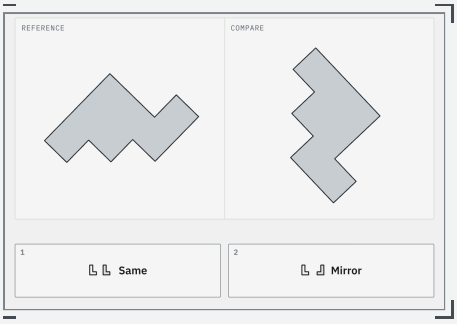
Angle	20° to 180°, in 20° steps
Axis	vertical → horizontal → in the picture plane → oblique
Cubes	6, 8, 10 or 12

STRATEGIES **Turn it as one piece:** picture the whole figure turning until it lines up; all arms in place means same. Quick for small and medium turns. **Follow one arm, then the next:** find an end arm in both figures and walk along it, checking each bend; the first bend that turns the other way means mirror. Steady for big or slanted turns. SWITCH when the turning picture blurs, follow arms; when following arms gets slow, turn the whole figure for a quick check. After two misses in a row, try the other way.

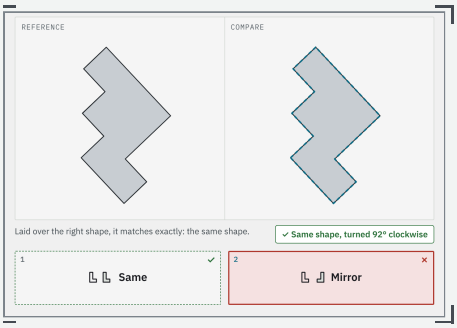
EVIDENCE Rotation and paper-folding practice transferred to each other, beyond a verbal control task, and held for new items (Wright et al., 2008). People who switch strategies more score higher on rotation tests (r = .50; Nazareth et al.), and checking a figure's overall shape predicts accuracy while counting cubes predicts errors (Hegarty, 2018).

EX-03 · Spin · SR

Spin 2D speeded rotation



QUESTION Same or mirror · a turn near 90° · 7 squares.



SHOW ME Turned 92° clockwise, the reference lies exactly over the right shape.

TRAINS Speeded rotation (SR): turning flat shapes quickly and accurately. Intrinsic-dynamic.

HOW TO PLAY Flat shapes made of squares. **Same or mirror:** is the right shape the left one turned flat on the screen, or its mirror image? **Pick two:** pick the two turned copies among four. The angles do not tell copies from mirrors.

DIFFICULTY GROWS WITH

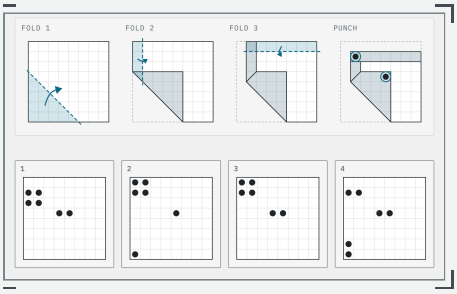
Angle	10° to 170°, in 20° steps
Squares	5 to 9
Time	none in Train; Speed mode and a deadline staircase in Threshold add it later

STRATEGIES **Turn it as one piece:** picture a cut-out card turning on the table until it lines up. Quick for small turns and simple shapes. **Check one feature, then its side:** find a notch or the longest arm in both shapes, walk round the outline the same way, and see which side the next part is on. Steady for big turns. **SWITCH** when the picture blurs, check features; when that gets slow, turn the whole card. Never count squares: a mirror image has the same ones.

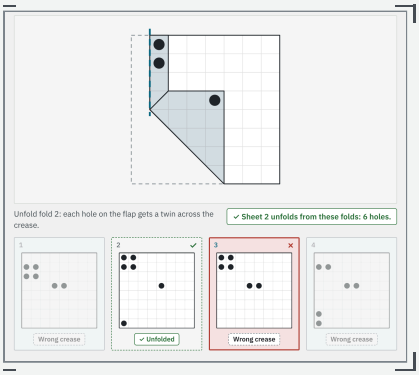
EVIDENCE Practice speeds rotation up a lot: over 21 days, response times fell from 3,132 to 1,480 ms while errors stayed flat (Wright et al., 2008). Across training studies, rotation response times improved more than rotation accuracy (g = 0.69 against 0.31; Uttal et al., 2013).

EX-04 · Fold · Vz

Fold Paper folding and hole punching



QUESTION Three folds, a corner fold first · two punches.



SHOW ME, PAUSED Unfolding fold 2: each hole on the flap gets a twin across the crease.

TRAINS Visualisation (Vz): following paper through folds and back. Intrinsic-dynamic.

HOW TO PLAY One panel per fold: the dashed line is the crease, the shaded part the flap, and the arrow shows where it goes. The last panel shows the punch, which goes through every layer under it. Pick the sheet you get when you unfold it (keys 1–5). All the options have the same number of holes.

DIFFICULTY GROWS WITH

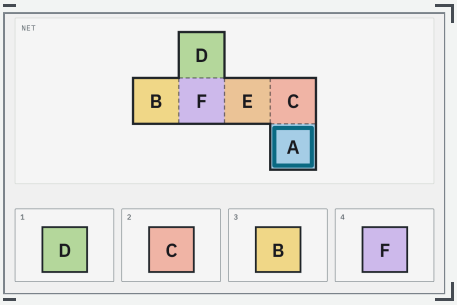
Folds	1 to 5
Fold type	half → corner → diagonal → mixed
Punches	1, 2 or 3
Idle folds	0 to 2 folds that add no holes; they make an item easier

STRATEGIES **Unfold it in your mind:** undo the folds, the last one first, and watch each hole on a flap get a twin across its crease. Quick for few, large flaps. **Follow one hole at a time:** for each punch, find the flaps it went through and mirror it across their creases, the last fold first. Steady for corner and diagonal folds. **SWITCH** when the flaps get crowded, follow one hole; when that gets slow, unfold the whole sheet. Check near-identical options hole by hole.

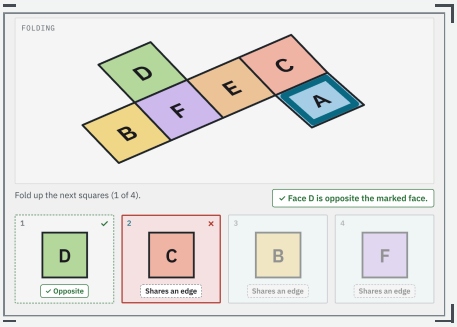
EVIDENCE Most people solve paper-folding tests by counting ("two holes per fold") rather than unfolding: only about 22 % mainly unfold, and the folds that matter predict difficulty better than the total (Burte et al., 2018). That is why every option here has the same number of holes. Off-screen challenge: fold and punch a real sheet.

EX-05 · Nets · Vz

Nets Cube nets



QUESTION Which face ends up opposite the marked face A? A 1-4-1 net.



SHOW ME, PAUSED The net folds up, one hinge at a time.

TRAINS Visualisation (Vz): folding a flat pattern into a cube in your head. Intrinsic-dynamic.

HOW TO PLAY Six squares joined at their edges; dashed lines are folds. Three questions: which face ends up **opposite** the marked face; do the two thick **edges** meet; does this pattern fold into a closed **cube**, every square on a side of its own? Keys 1–4, or 1 = Yes and 2 = No.

DIFFICULTY GROWS WITH

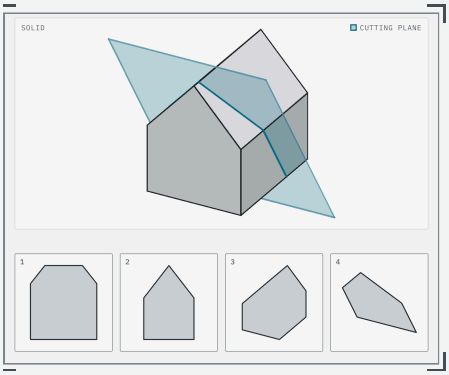
Carried	squares that move with each fold: from 2–4 up to 11–15
Net shape	1-4-1 → 2-3-1 → staircase nets (2-2-2, 3-3) with no long row
Between	2 or 3 faces between the two in question (opposite)

STRATEGIES **Fold it in your mind:** keep one square flat and fold the others up one hinge at a time, carrying the squares attached to each. Works on every net. **Use what nets always do:** joined squares are never opposite; in a row of three the ends are opposite; edges that form an inside corner always meet. Fast where a net has rows and corners. **SWITCH** use a fact when one fits, and fold when none does. After a miss, Show me folds the net.

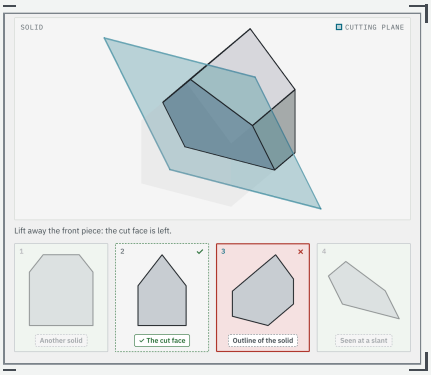
EVIDENCE The time to fold a net mentally grows with the number of squares carried along (Shepard & Feng, 1972), so that count sets the difficulty here. A cube has only 11 nets, so "does it fold into a cube" can be learned by heart; the questions about faces and edges need the fold itself. Off-screen challenge: cut out and fold a net.

EX-06 · Slice · Vz

Slice Cross-sections



QUESTION Two joined solids · an oblique cut.



SHOW ME, PAUSED The front piece lifts away and leaves the cut face; tags name each outline.

TRAINS Visualisation (Vz): imagining the face that a cut leaves. Intrinsic-dynamic.

HOW TO PLAY A tinted plane cuts through a solid: one solid, two joined, or one inside a see-through other. Which of four outlines is the cut face, turned to face you? A grey region inside an outline is the inner solid's part of the cut.

DIFFICULTY GROWS WITH

Structure	simple → joined → embedded
Cut	orthogonal → oblique → double oblique
Curved	flat-faced solids, then cylinders and cones

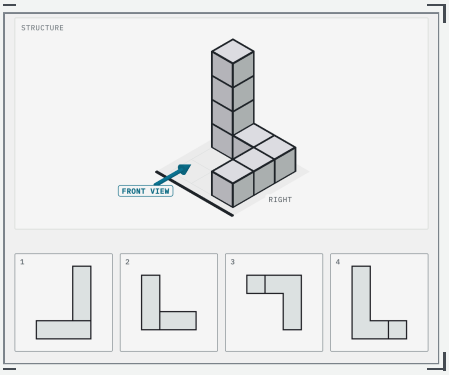
STRATEGIES **Slice it and turn the face:** lift away the piece in front of the plane and turn the cut face toward you. Quick for single solids and straight cuts. **Walk around the cut:** every flat face the plane crosses adds a straight edge, a curved surface adds a curve; count them. Steady for tilted cuts and solids inside solids. **SWITCH** when the picture blurs, walk round the edges; when counting gets slow, turn the face for a quick look. Never pick the shape as it looks from here: seen at a slant, the face looks squashed.

EVIDENCE In a cross-section test, tilted cuts were harder at every level of structure, and the most chosen wrong answer was the shape seen from the viewer's own position instead of the plane's (Cohen & Hegarty, 2007); Parallax always includes that option. Short interactive cross-section training helped low-spatial students, in a second experiment also on untrained figures (Cohen & Hegarty, 2014).

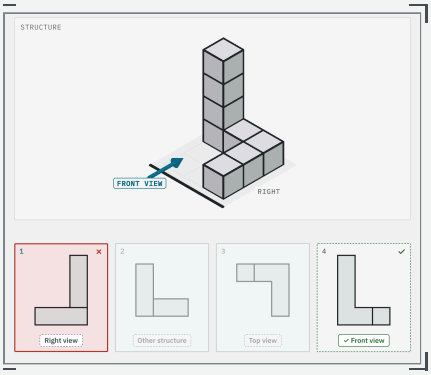
EX-07 · Views · Vz

Views

Orthographic views of block structures



QUESTION The front view · 8 cubes on a 3 × 3 grid.



FEEDBACK Tags name each wrong drawing: the right view, another structure, the top view.

TRAINS Visualisation (Vz): matching a 3D structure with its flat views, as engineering drawings do. Intrinsic-static.

HOW TO PLAY Unit cubes stacked on a grid, seen from the front-right corner; the thick edge of the base is the front. Three questions: which line drawing shows the structure from the side the arrow marks; which structure has exactly these top, front and right views; how many cubes are there? Every cube shows at least one face.

DIFFICULTY GROWS WITH

Cubes	4, 6, 8, 10 or 12
Grid	3 × 3 or 4 × 4
View	top → front → right

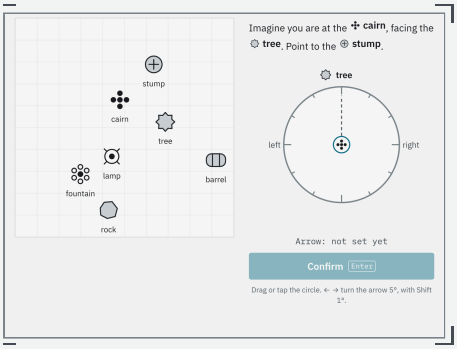
STRATEGIES **Walk round it:** move in your mind to the marked side and look straight on, so that depth flattens out. Quick for small structures and the top view. **Read it column by column:** note each column's height on a top view; the front view shows the tallest stack in each column, the right view the same row by row, front on the left. Steady for large structures and side views. **SWITCH** when the picture slips, use the heights; when counting gets slow, picture the view. A mirrored drawing has the same parts in the opposite order: check which side the front is on.

EVIDENCE Orthographic and isometric views are core modules of a spatial-skills course for first-year engineering students. Scores on a rotation test rose from about 50 % to 74–82 % correct, but these students were selected for low scores, so part of the rise is regression to the mean; the numbers are before and after, and the comparison groups were self-selected, so causal claims are weak (Sorby, 2009; Sorby & Veurink, n.d.). Off-screen challenge: build a structure, then sketch its views.

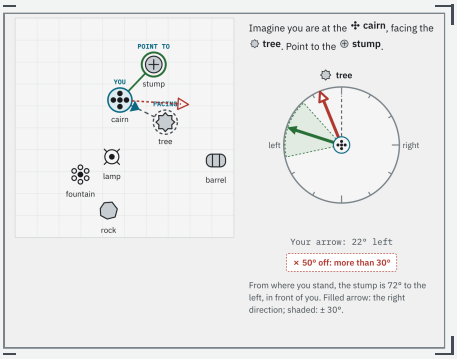
EX-08 · Orient · PT

Orient

Perspective taking



QUESTION Stand at the cairn, face the tree, point to the stump.



FEEDBACK The map marks you, what you face and the target; the filled arrow is the right direction.

TRAINS Perspective taking (PT): imagining a scene from somewhere else. Extrinsic-dynamic.

HOW TO PLAY A map shows objects from above. Imagine standing at one object, facing a second: which way is the third? Set the arrow on the circle (drag or tap, or ← and → in 5° steps, with Shift 1°), then confirm. The top of the circle is the object you face, not the top of the screen.

DIFFICULTY GROWS WITH

Shift	the imagined turn: 0–45° → 45–90° → 90–135° → 135–180°
Objects	5, 7 or 9
Tolerance	within 40°, 30° or 20° counts as right

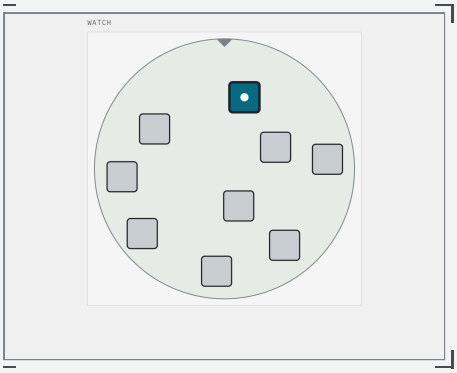
STRATEGIES **Step into the map:** imagine standing at the first object and turning on the spot until you face the second, then point. Natural for small turns. **Measure the angle:** estimate the angle between the line to what you face and the line to the target; clockwise is right. Steady when you face down the map. **SWITCH** if you catch yourself pointing as if the top of the screen were ahead, step into the map again; for large turns, measure. After a miss of more than 90°, decide front or behind first.

EVIDENCE Perspective taking is separate from mental rotation, though strongly related to it (Hegarty & Waller, 2004). Errors grow with the imagined turn (r = .50; He et al., 2022), and beyond about 90° people tend to fall back on their own body's frame (Kozhevnikov & Hegarty, 2001). Rotation training transferred to perspective taking in one study (Meneghetti et al., 2016); our literature search found no trials that trained perspective taking directly.

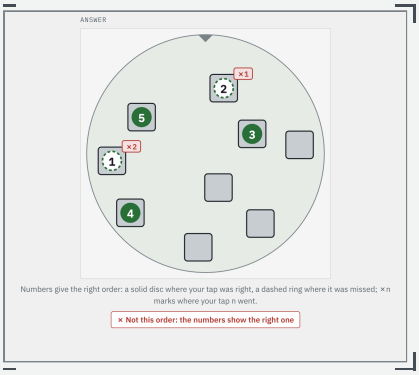
EX-09 · Span · MV WM

Span

Block span: a sequence of places



QUESTION A sequence of 5 blocks lights up, about one a second.



FEEDBACK Numbers give the right order; ×1 and ×2 mark where taps 1 and 2 went.

TRAINS Visual memory (MV) and spatial working memory (WM): holding a sequence of places in mind. Outside the 2 × 2.

HOW TO PLAY Nine blocks on a board light up one at a time. Then tap them in the same order (**forward**) or in reverse, the last one first (**backward**). Backspace takes back a tap. At some levels the board turns before your turn; the blocks travel with it, so tap them where they are now.

DIFFICULTY GROWS WITH

Length	2 to 9 blocks
Crossings	a path that crosses itself: none, some or many times
Board turn	0°, 90° or 180° between showing and recall

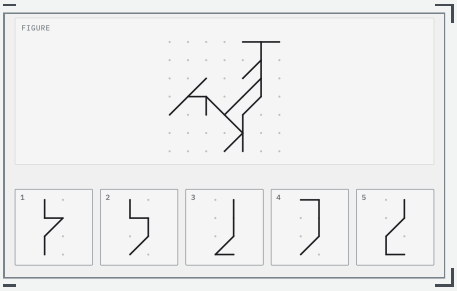
STRATEGIES **See the path as a drawing:** join the blocks with an imagined line as they light, and remember the shape it makes. Good for paths that do not cross. **Group and repeat:** split the sequence into groups of two or three, give each block a place name, and repeat the groups to yourself. Steady for long or crossing paths. **SWITCH** when the drawing tangles, group; when the groups blur, look for the drawing. With a turn, turn the whole drawing with the board.

EVIDENCE Path crossings lower recall, while spatial structure and grouping help (De Lillo et al., 2016). Working-memory training gives near transfer to other working-memory tests but no convincing far transfer (Melby-Lervåg et al., 2016). **Test hygiene:** block-tapping practice can raise scores on clinical block-tapping tests, so mention your practice if you take one.

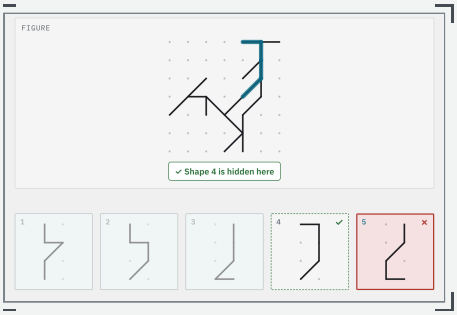
EX-10 · Hidden · CF

Hidden

Hidden shapes in a line figure



QUESTION Which of five shapes is hidden? 14 extra lines.



FEEDBACK Shape 4 is drawn in where it hides.

TRAINS Flexibility of closure (CF): finding a known shape in a busy pattern. Intrinsic-static.

HOW TO PLAY A figure made of short lines between the dots of a grid, and five small shapes at the same size. Exactly one shape is hidden in the figure, the same way up: every line of the shape must be a line of the figure. Lines of the figure may run on past the shape's ends, and others may touch it.

DIFFICULTY GROWS WITH

Extra lines	6, 10, 14, 19, 25 or 32
Similarity	low → medium → high: the hardest options differ by one line
Size	a figure 5, 6 or 7 grid cells wide

STRATEGIES **Slide the shape over the figure:** hold it as one picture and move it over the figure until every line lands on a line. Quick for small or sparse figures. **Anchor on one part:** pick the shape's most unusual part, such as a slant or a tight corner, find each place it occurs, and check the rest there. Steady for busy figures. **SWITCH** if the whole shape keeps getting lost, anchor; if anchoring gets slow, look for the whole. Lines running past a shape are allowed: do not let them hide it.

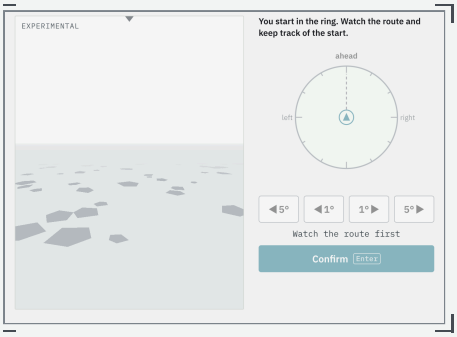
EVIDENCE Embedded figures are the classic intrinsic-static task (Uttal et al., 2013); spatial training raised young children's scores on tasks of this kind (g = 0.46; Yang et al., 2020). Our literature search found no studies of training it in adults. A full search proves that the right shape fits in exactly one place and that no other option fits at all.

EX-11 · Homing · NAV

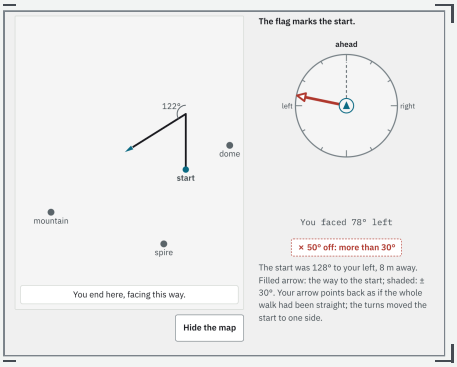
Homing

Navigation with landmarks

experimental



QUESTION On the walk: the circle and the turn buttons wait until the route ends.



SHOW ME, PAUSED The route from above, with its 122° turn and three landmarks; a hint names the slip.

TRAINS Navigation (NAV): keeping track of where you started as you move, with far landmarks as fixes on the easier levels. Extrinsic-dynamic. **Experimental.** It is the landmark-aided stage before Reckon.

HOW TO PLAY You watch a short walk through your own eyes: straight legs, with turns on the spot between them. You start in a ring that disappears once you walk away. Then turn to face the start: drag the view sideways, drag or tap the arrow circle, or use the buttons for 5° and 1° steps (keys ← and →, Shift for 1°), and confirm. Far landmarks barely move as you walk, so, like a navigator's fix, they show how far you have turned.

DIFFICULTY GROWS WITH

Route	2 to 4 legs with small (30–60°), medium (60–120°) or large (120–160°) turns, in eight steps
Landmarks	three → one → none
Tolerance	within 40°, 30° or 20° counts as right

STRATEGIES **Draw the map in your head:** picture the walk from above as it happens, with yourself as a moving arrow. Natural for short routes. **Track the start like a compass needle:** keep one direction in mind, where the start lies from you, and update it after every leg and turn. Steady without landmarks. SWITCH when the picture blurs after several turns, track the direction; with landmarks, check your picture against them.

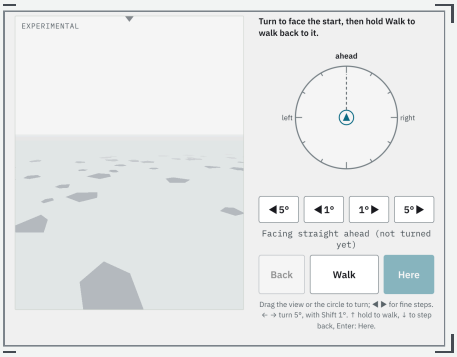
EVIDENCE On a screen, Homing measures visual path integration only, without the body's own sense of turning and walking; whether it helps real wayfinding has not been studied. Since 1.1 it walks at 1.8 m/s and turns at 45°/s with gentle starts and stops, over routes of 12 to 28 m (page 19).

EX-12 · Reckon · NAV

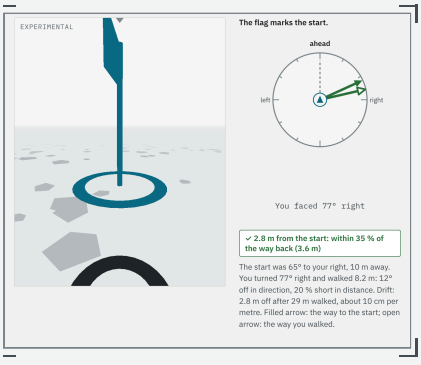
Reckon

Dead reckoning, nothing to steer by

experimental



QUESTION After the walk nothing marks the start. Turn, hold Walk, then tap Here.



FEEDBACK A walk back that ended near the start: the errors in direction and distance, and the drift per metre walked.

TRAINS Navigation (NAV): path integration by dead reckoning, keeping track of home from your own turns and steps alone. Extrinsic-dynamic. **Experimental;** rated per screen size.

HOW TO PLAY Watch a walk of 2 to 4 legs across an endless plain. **Point:** turn to face the start and confirm. **Return:** turn, then hold Walk (↑) to walk back, Back (↓) to step back, and tap Here (Enter) where the start is. Nothing in the world, the sky or the screen marks a place; above the first speed level, speeds vary, so seconds do not measure distance (page 19).

DIFFICULTY GROWS WITH

Route	Homing's eight steps
Ground	textured → sparse → short-lived dots that fade within about a second
Speed	steady, about 1.8 m/s → varied between legs (0.5 to 2.0 m/s) → within legs too
Tolerance	±40°, 30° or 20°; on return items within 50, 35 or 25 % of the way back

STRATEGIES **Picture the shape from above,** most accurate on longer routes, or **keep the start as a direction from you,** updated with every turn and leg, quick when the turns come fast. SWITCH after two misses in a row.

EVIDENCE In a lab version of the task with feedback after every trial, the overshoot of distance vanished within one session and the angular bias shrank, though practice and feedback were not separated (Noel et al., 2020). Whether gains last, or reach walking in real places, has not been studied.

Navigation without fixes

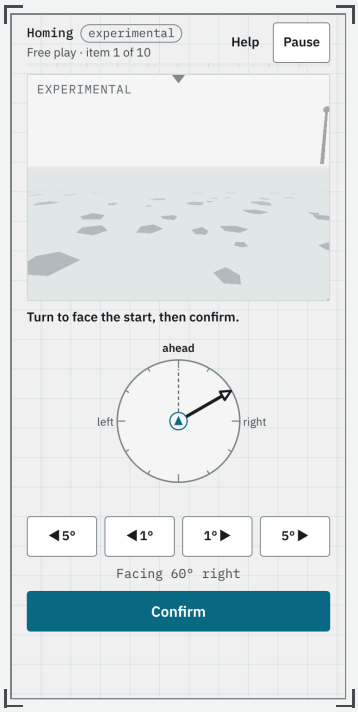


FIG. 3.2 Homing on a phone, turned 60° right; a far landmark stands at the right edge.

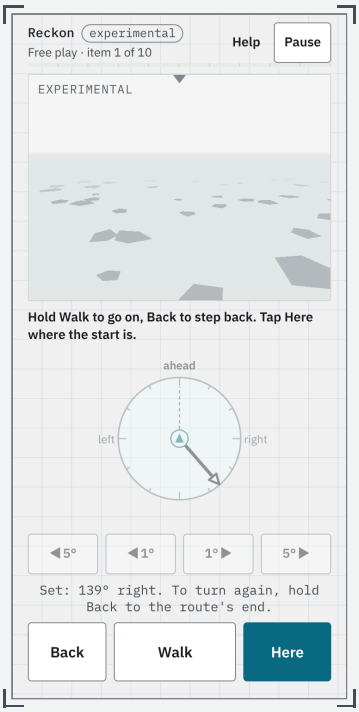


FIG. 3.3 Reckon on a phone, part of the way back.

Why Reckon has no landmarks

Dead reckoning works out where you are from where you started and every turn and distance since. Its errors add up, so navigators correct them with fixes: a landmark, a star, a satellite. A community member pointed out that Homing's far landmarks are exactly such fixes, so they correct the very errors that path-integration training should work on. Reckon removes every fix:

- no objects, a flat sky without a sun, unlit ground;
- ground marks laid out anew for every item: they show turning and walking, but mark no place;
- a start ring that is gone once you leave it;
- no map, compass or counter during the walk;
- above the first speed level, speeds that change, so seconds do not measure distance.

Homing keeps its landmarks as the easier stage before Reckon.

Feedback that trains the error

A Reckon answer is right within the tolerance, and only that moves your rating. The feedback says more: the angle error; on return items the distance error and the drift, how far off you ended per metre walked; a hint when an answer matches a typical slip (turns left out, left and right swapped, turning straight round, a heading never updated); and, after five routes, your habit over the last 20, turning too far or too little and walking short or long, with a chart. Harder items bring tighter tolerances. The trial log keeps the signed angle error and, on walks back, the distance walked as a share of the true one, so pooled data can tell the two apart.

Answers that need no tracking (the turns reversed, a heading never updated, the mirror image) always miss, and turning straight round or giving one fixed answer stays near chance wherever the routes allow. As the Library says: "On the easier routes the turns alone tell you much of the way home; distances matter more with bigger turns, at tighter tolerances and when you walk back." Routes run 12 to 28 m, the start lies 5 to 20 m away, and no walk moves for more than 30 seconds.

Touch, and comfort

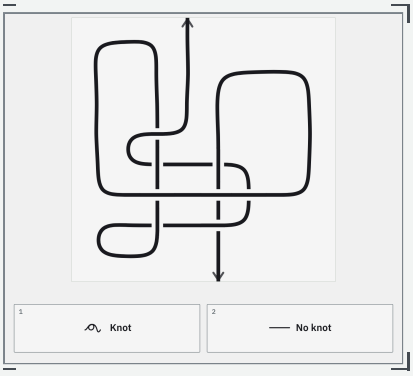
Both exercises play by touch: drag the view to turn, then fine-tune with the 1° and 5° buttons; captions sit over the view as subtitles, and held sideways the view and the controls sit side by side. Pause freezes the walk at once. The motion is gentle: Homing walks at 1.8 m/s; Reckon too on its steady level, and between 0.5 and 2.0 m/s above it, so a leg's walking time says little about its length. Turns run at 45°/s at most, with soft starts and stops. If the motion makes you feel unwell, pause or stop.

A trial is voided only when the walk itself was harmed: a freeze of more than a second while moving, slow frames for more than half of it, a lost 3D view, the screen changing shape during the walk or while you answer, or a pause. Homing's items changed in 1.1, so its new trials are logged as version 2.

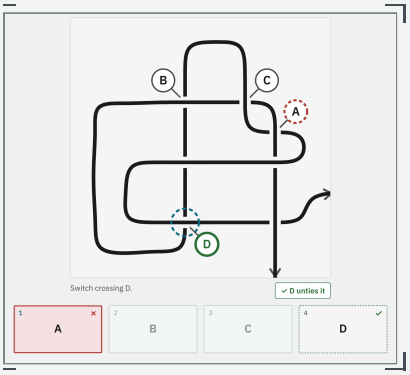
EX-13 · Knots · Vz

Knots

Knots and links



QUESTION Pull · 8 crossings, open spacing: does a knot remain?



SHOW ME, PAUSED Untie: switch crossing D, and the moves take the rope apart.

TRAINS Visualisation (Vz): following a rope through its crossings and picturing it pulled tight. Intrinsic-dynamic.

HOW TO PLAY One rope, its ends running off the plate. It is perfectly slippery and cannot pass through itself; where two strands cross, the gap shows which one is underneath. Four formats share one rating: **Pull**, does a knot remain when both ends are pulled (1 Knot, 2 No knot)? **Untie**, which of four marked crossings, switched, lets the rope fall open (tap a letter or press 1–4)? **Tie**, which one ties a knot? **Link**, are two closed loops linked, or could they be pulled apart?

DIFFICULTY GROWS WITH

Rope	5 to 12 crossings in ten steps; on most higher steps, plain ropes that need a slide-over before anything comes off
Spacing	open → close → tight strands
Format	pull → untie and tie (7 crossings or more) → link (8 to 11, the loops crossing six times)

STRATEGIES **Picture the pull:** hold one end still and pull the other in your mind; watch which loops slide off and which tighten. Quick on short ropes. **Trace and remove:** take the rope apart move by move, as Show me does; if what is left goes over, under, over all the way round, it is a knot. Steady on long or crowded ropes.
SWITCH when strands run close, trace; when tracing gets long, picture the whole rope pulled tight. Counting crossings never decides: a knot and a plain rope of the same shape are both here.

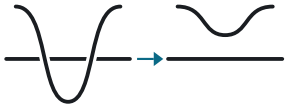
EVIDENCE The one psychometric knot test, the Knot Reasoning Task, correlated r = .54 with mental rotation and r = .59 with paper folding (Bennett-Pierre et al., 2025). People recognise a knot's shape far better than they judge its strength (Croom & Firestone, 2024), so Knots asks only about shape. No study of training knot diagrams was found, and nothing shows that it helps tie real knots.

How every item is proved

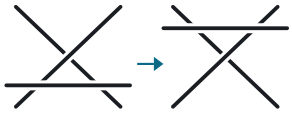
A rope is a curve through points on a grid, with a flag per crossing for the strand on top. Before you see an item, a validator finds the crossings again from the drawn curve and settles the answer in up to three steps, each a proof. The three moves never change whether a knot is there.



UNTWIST A CURL A loop that crosses only itself comes out.



LIFT OFF A LOOP On top at both of its crossings, a loop slides off.



SLIDE OVER A CROSSING A strand on top of both strands slides across.

1 · Take it apart. Apply the moves wherever they remove crossings. If every crossing comes off, there is no knot. Show me replays these moves, checking each one, and stops with a caption where one cannot be drawn safely.

2 · A determinant. If crossings remain, compute the diagram's determinant (by Fox colouring). A rope without a knot always gives 1, so any other number proves a knot.

3 · The Jones polynomial. Otherwise slide-overs and the moves again, then the Jones polynomial, which for up to 24 crossings equals 1 only without a knot (Tuzun & Sikora); ropes here have at most 12.

Untie and tie: switched, exactly one of the four marked crossings changes the answer. **Linked** loops are proved by a linking number other than zero or by the Jones polynomial; **apart** only by moves that leave no crossing between them.

No cheap tells. Each knot is paired with a plain rope from the same outline, matched on over-under alternation, twist and the moves in plain sight. Unpaired, alternation alone told knots apart 75 to 92 % of the time; paired, no cheap rule tested beat 52.3 % on Pull, where guessing scores 50 %, or 33 % on Untie and Tie, where it scores 25 %.

DAT-style formats EX-14 to EX-19

Six exercises practise the six question formats of the Perceptual Ability Test (PAT), the perceptual part of the Dental Admission Test (DAT®), in the test's order. Every item is generated by Parallax and checked by a solver like every other exercise, and the rules are written in Parallax's own words.

PART	IN THE TEST	EXERCISE	WHAT YOU DO	ANSWERS
1	apertures	EX-14 Openings	find the opening a pictured solid passes through, turned the right way, with nothing to spare	5 openings
2	view recognition	EX-15 Missing View	two of the top, front and end views are given: pick the third, hidden edges dotted	4 drawings
3	angle discrimination	EX-16 Angles	rank four angles from smallest to largest	4 orderings
4	paper folding	EX-17 Hole Punch	a 4 × 4 sheet folded two or three times and punched once: pick the unfolded sheet	5 sheets
5	cube counting	EX-18 Painted Cubes	how many cubes of a painted figure have exactly so many painted sides	1 to 5 cubes
6	form development	EX-19 Form	which of four solids a flat pattern folds into	4 solids

The DAT-style focus, opt-in

Off by default. Off, the six formats stay in the Library, Free play and Section practice, and never come up in Train, Today's plan or the Daily Challenge. Switch the focus on, in the Library's DAT-style group or in Settings, when you are preparing for the test: they then join Train and Today's plan like any exercise. Every switch is kept in your record and its backup, so pooled data can tell practice with and without the focus apart.

Why opt-in

Practising a test's formats raises scores on those formats. Leaving them out of default training keeps the choice with you, and keeps the Spatial Check clean: no format here puts oriented marks on cubes, which is the Cube Match probe's format. "Every exercise" and "Broad base" do not count these six.

Angles is perceptual-judgement practice

Like Track's attention-practice label: judging angles by eye improves with practice, but the gains tend to stay with this kind of figure. Once you have practised Angles since your starting checks, Progress adds a note to compare the Level probe with care, since judging angles by eye may carry over a little to it.

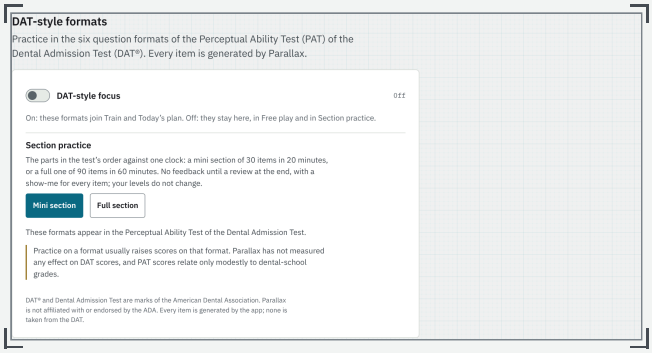


FIG. 3.4 The Library's DAT-style group: the focus switch, Section practice (page 28) and the limitation line.

WHAT PRACTICE HERE CAN AND CANNOT DO

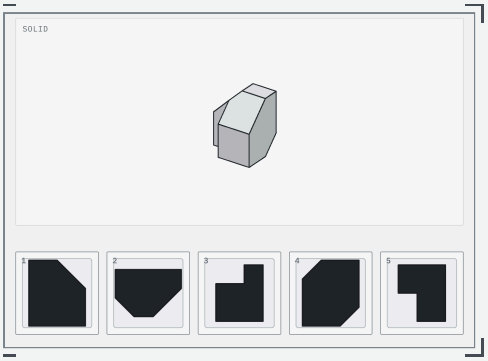
Practice on a format usually raises scores on that format: near transfer, in this guide's terms. Parallax has not measured any effect on scores in the test itself, and predicts no score. In the ADA's own validity study, PAT scores correlated .16 to .36 with dental-school grades after corrections, highest for second-year technique courses (American Dental Association, 2026). The project's search found no study that trained these formats and then measured the test against a control group.

DAT® and Dental Admission Test are marks of the American Dental Association. Parallax is not affiliated with or endorsed by the ADA. Every item is generated by the app; none is taken from the DAT.

EX-14 · Openings · Vz

Openings

Which opening fits the solid



QUESTION Features 2 + 1 + 1 with slanted cuts; the fitting face may be any of the three. Each opening is turned or flipped at random.

HOW TO PLAY The solid may be turned any way before it goes in, then slides straight through, so the opening it needs is its outline seen along that direction. The openings are drawn at the solid's scale and must fit exactly: the right features in the wrong proportions are wrong. Every cut runs straight through the solid.

DIFFICULTY GROWS WITH

Features	per face: 1 + 1 + 1 → 2 + 1 + 1 → 2 + 2 + 1 → 2 + 2 + 2
Edges	straight → slanted (45°) cuts
Fitting face	mostly the front → any face → mostly the top; upright → lying
Wrong ones	other features → also one outline stretched or squeezed

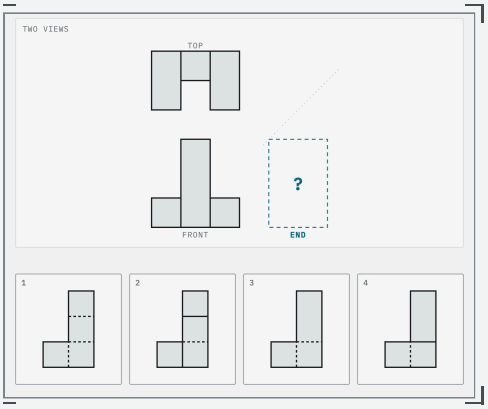
STRATEGIES **Push it through:** pick a face and picture the hole it would cut. **Three outlines, then sizes:** the solid has only three; match width and height, then the features, then the lengths.

CHECKED Rays cast through the solid on a grid of twelfths of a unit rebuild its three outlines; every opening is compared under all eight turns and flips, and exactly one matches.

EX-15 · Missing View · Vz

Missing View

Two views given, pick the third



QUESTION Top and front given: which drawing is the end view? All four share one outline; only the lines differ.

HOW TO PLAY Top (upper left), front (lower left) and end (lower right) always sit in the same places, at one scale; two are drawn. The end view looks at the right side, so the front is on its left. A solid line is an edge you can see, a dotted line one hidden behind nearer cubes. Every cube stands on the ground or on another cube.

DIFFICULTY GROWS WITH

Cubes	4–5 → 6–7 → 8–9 → 10–12, on a 3 × 3 or 4 × 4 grid
Missing	end → front → top view
Wrong ones	another outline → the same outline, other lines → one edge solid instead of dotted, or the reverse

STRATEGIES **Build it, then look** from the missing side, or **line up and eliminate:** every line of the missing view lines up with lines in the given ones.

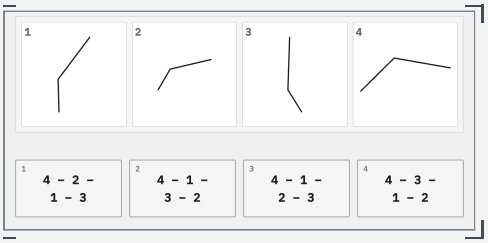
CHECKED Two views do not fix a structure, so every item searches all stacks of cubes that fit the two given views, and a wrong drawing is admitted only when none of them has it: exactly one drawing is possible. A second search, built on the line drawing alone, must find the same stacks.

EX-16 · Angles · LE

Angles

Angle ranking

perceptual-judgement practice



QUESTION Four wide angles, turned freely: which list runs from smallest to largest?

HOW TO PLAY Four numbered angles; each answer lists the numbers from the smallest angle to the largest. Only the opening between the arms counts: arm length and how an angle is turned do not. Every angle is less than a straight line.

DIFFICULTY GROWS WITH

Closest gap	6° → 5° → 4° → 3° → 2°
Size	near right angles → sharp → wide
Deciding pair	the smallest and largest → the middle pair → the closest pair
Pose	one arm level → turned freely

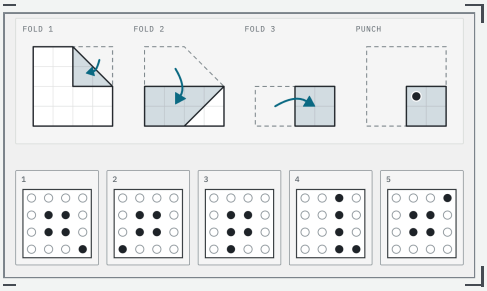
STRATEGIES **Settle the extremes first** and cross out every list that disagrees; often the close pair never needs judging. **Lay one on another:** corners and one arm lined up, the wider angle sticks out.

CHECKED Judging angles by eye is most precise near 90° (Dillon et al., 2019). Arm lengths are drawn so that quick cues, such as the distance between the tips, the area between the arms or arm length, point to random lists, and rules that read only the lists stay near chance; each angle is measured again from the drawn ends.

EX-17 · Hole Punch · Vz

Hole Punch

Fold, punch once, unfold



QUESTION Three folds, then one punch through every layer under it: which sheet?

HOW TO PLAY A square sheet of 4 × 4 cells is folded two or three times: the dashed outline is where the paper was, the shaded flap is the part that just moved. Then one punch goes through every layer under it. Pick the unfolded sheet: a filled circle is a hole, an open ring none. The paper is never turned over and never leaves its square; all five sheets have as many holes.

DIFFICULTY GROWS WITH

Folds	2 → 3
Fold type	half → corner → diagonal → mixed
Idle folds	0 to 2 folds the punch misses; they make an item easier

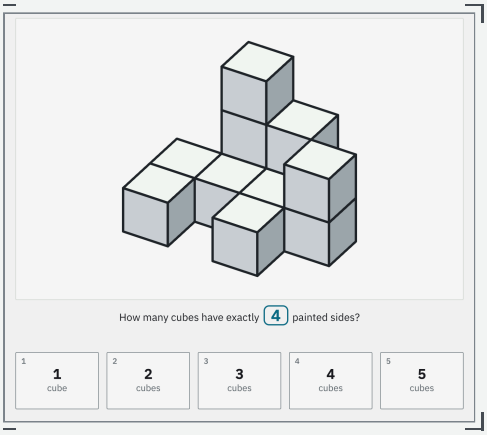
STRATEGIES **Unfold it in your mind,** the last fold first: every hole on a flap gets a twin across its crease. **Track the hole on the grid:** mark the punch, then mirror it across each crease with paper under the punch.

CHECKED Fold's exact paper model on a smaller sheet; an independent re-simulation checks every drawn step. Unfolding backwards by symmetry about the creases, the format's own method, finds most answers; partial flaps and mixed folds are where the layers must be tracked.

EX-18 · Painted Cubes · Vz

Painted Cubes

Count by painted sides



QUESTION How many cubes have exactly 4 painted sides?

HOW TO PLAY Equal cubes glued together, painted on every outside face except those resting on the ground. How many cubes have exactly the number of painted sides asked? A cube out of sight is there only to hold up cubes above it, and every column's top cube is in view.

DIFFICULTY GROWS WITH

Cubes	6–9 → 10–13 → 14–17 → 18–22
Tallest column	2 → 3 → 4 cubes
Sides asked	3 or 4 → 1 or 2, which sit inside the figure or under other cubes

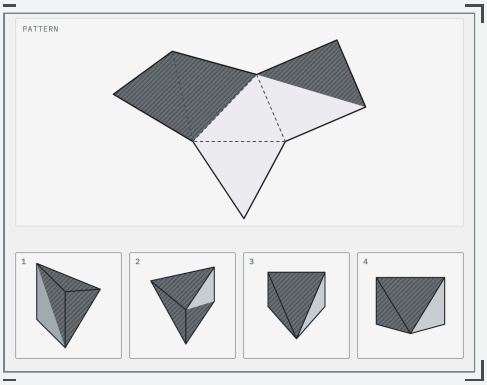
STRATEGIES **Know the common positions:** a corner cube on the ground has 3. **Tally column by column,** hidden cubes included; tall columns call for the tally.

CHECKED Five painted sides is never asked: such cubes are rare, so the answer would give itself away. The count is drawn evenly from its range, and the classic error, counting the painted faces in view (Kurtuluş & Yolcu, 2013), is held near chance. Section practice asks up to three questions per figure.

EX-19 · Form · Vz

Form

Pattern folding



QUESTION Two shaded faces and a half-shaded one; four wedges on their sides, each turned its own way.

HOW TO PLAY The pattern is a solid's outside laid flat: dashed lines fold; solid lines are cut edges that meet another edge once folded. Shaded faces carry a tone and a hatch; a half-shaded face also shows which way it turns. Pick the solid the pattern folds into, marks in the right places; only faces turned toward you count.

DIFFICULTY GROWS WITH

Solid	boxes, prisms → wedges, pyramids → frusta, five-sided prisms
Marks	shaded and half → two shaded and half → shaded and two halves
Pictures	upright → on a side → upside down
Wrong ones	a size off, or marks moved → marks moved only

STRATEGIES **Fold it up** around one face kept flat, or work by **landmarks:** rule out by shape, then check one marked face and its neighbours.

CHECKED Folded in 3D, the pattern must close and match exactly one picture under some rotation; a second check that never sees the solid lays its pieces onto each picture face by face, and must agree. Marks move only to faces of the same shape and size. Never a cube: oriented marks on cubes belong to the Cube Match probe.

CHAPTER 04

Progression modes

Parallax has five ways to practise and one to measure, and they share one record of your learning. There are several because good ideas about learning genuinely conflict: a single mode would have to pick a side and lose what the other side gets right.

Four conflicts that cannot be one setting

CONFLICT	WHY BOTH SIDES HAVE A POINT	PARALLAX'S ANSWER
Mastery gates or free practice	Gates keep each step secure before the next. Freedom supports autonomy, which predicts enjoyment (Ryan et al., 2006). One restricts what you can reach; the other removes the restriction.	**Path** is gated, **Free play** is open, and **Train** sits between: no gates, but the engine picks the difficulty.
Training or measuring	Training should adapt and explain. Honest measurement needs a fixed protocol, no feedback, and tasks that training never touches.	Train, Path and Speed train. The **Spatial Check** measures, and training never feeds it.
Untimed or timed	Accuracy is built without a clock; fluency needs one. Deadlines change the best strategy and, reportedly, what rotation tests measure (Voyer, 2011).	Train is untimed. **Speed** opens later, on settings you already answer well, with its own timed rating.
A standard threshold or a training target	Research protocols use staircases that settle where half the trials are right; training aims higher, at 70 to 85 % right depending on the format (Wilson et al., 2019; Pelánek et al., 2017). The two numbers cannot be compared.	**Threshold** keeps the research protocol for Track as a Standard preset; Train aims at your own target.

The modes, and the Daily Challenge

MODE	FOR	HOW IT WORKS	YOUR RATING
Train default	everyday practice	Three exercises per session (focus, review, variety); items aim at your style's target; five interleaving stages, from blocked practice to fully mixed.	updated
Path	beginners; steady steps	Ten stages per exercise, one at a time. A stage passes when your recent answers and your rating agree; passed stages come back for rechecks. Never says "mastered".	updated
Threshold	measuring one setting	One setting moves on a staircase while the rest stay fixed: Track Standard, Span length, Spin deadline.	untouched; results kept apart
Speed	fluency	Deadlines on settings you already answer well without a clock.	its own timed rating
Free play	choice	Pick the exercise, variant, settings and timer. Logged.	untouched
Spatial Check	measurement	Short adaptive tests on tasks you never train, without feedback (chapter 5).	untouched
Daily Challenge	the community	The same 8 items for everyone on each date (your local one), at medium difficulty, one scored attempt (replays are practice); an attempt left part-way goes on where it stopped. Track, Homing and Reckon are left out because their difficulty depends on the screen, and the six test formats because they are opt-in. The share line reads `Parallax Daily #N · x/8 · discord.gg/chmr`; #1 was 1 November 2026.	untouched
Section practice 1.1	rehearsing six test formats	The six formats of page 21 in the test's order, against one clock: 30 items in 20 minutes or 90 in 60, no feedback until a review at the end (page 28).	untouched

Friends can also share a challenge code to play exactly the same items and compare notes. There is no ranking anywhere.

The shared engine, in plain words

Your rating

For each exercise Parallax keeps a belief about your ability: a best guess and how sure it is. Every answer moves it, a lot while it is unsure and a little once it has settled. The uncertainty grows slightly with every answer and every day off, so the rating keeps up when you learn or forget. Stuttered, voided or skipped trials never count.

Predicted difficulty

Every setting adds a known step of difficulty (a wider angle, one more fold, two more cubes), so even a brand-new item has a predicted difficulty. This is a linear logistic test model (Fischer, 1973); in mental rotation, six item features reproduced test difficulty at r = .90 (Rajeb et al., 2025). The chance of a right answer is

```
P(right) = g + (1 − g − 0.03) · σ(θ − b)
```

where g is the guessing rate of the format, θ your ability, b the item's difficulty, σ the logistic curve and 0.03 a small lapse rate. Parallax's steps are set by hand for now (chapter 8).

Your style and target

Items aim at a success rate $p^\star = g + (1 - g) \times k$, with k set once by your style:

STYLE	K	4 OPTIONS	YES / NO	OPEN
Comfortable	0.85	89 %	93 %	85 %
Balanced (default)	0.70	78 %	85 %	70 %
Stretch	0.55	66 %	78 %	55 %

For two-choice tasks and simple learners, about 85 % right maximises learning in theory (Wilson et al., 2019); Balanced gives that on yes/no items. In a large adaptive-practice study, learners rated 60–70 % success as about right (Pelánek et al., 2017). In children's maths, higher success meant more practice, and that volume carried the gain (Jansen et al., 2013): hence Comfortable.

How a Train session is built

Three exercises: a focus, a review and a variety pick. They share the trials according to how fast each is improving (measured on the rating, because success itself is held at the target), with a bonus when a review is due and for rest. Each gets 20 to 50 % of the trials, and 15 % of picks are random. In children's maths practice, a pure learning-progress bandit reached higher levels than an expert sequence on three of four exercise types but starved the fourth (Clement et al., 2015), hence the floors. Untried exercises come first, so each is tried within the first few sessions (the six test formats only under their focus). Sessions open with two easy warm-up items and close on an easier one.

Levels 1 to 100

Your level is the difficulty you would answer right with probability g + (1 − g) × 0.75, placed on the exercise's own range: level 1 is its easiest item, level 100 its hardest. The reference is fixed, so changing your style never changes your level. Each level comes with a likely range (95 %) and the real settings at that difficulty, such as "120° turns, 10 cubes". It moves at the end of each block of 10 answers, not after every answer; block notes, summaries and bests all show this one level. Levels compare you with your own past on one exercise; they are not scores against other people.

Which item comes next

In every 10 items, 8 aim at your target, 1 is easier and 1 is a stretch. Among equally good candidates, Parallax prefers the settings and variants used least recently, and does not repeat the last three combinations while another one fits. Varying what does not decide the answer helps learning generalise (Raviv et al., 2022). Where you do clearly worse than predicted on one setting, over at least 15 answers, a fifth of target items come from it: a weak spot.

Interleaving stages

STAGE	RUNS OF	WHAT CHANGES
S1	20	one variant per run
S2	10	variants mixed
S3	5	shorter runs
S4	2	alternating with a similar-looking exercise, without its name shown
S5	random	about 70 % of trials switch exercise

An exercise moves up a stage once it has 40 answers and a settled rating, and back when trials right after a switch fall more than 10 points below prediction. Stretch starts at S3. Raising interference step by step beat both purely blocked and purely random practice (Porter & Magill, 2010), and interleaving helps most when categories look alike (Brunmair & Richter, 2019). A name banner would remove the choice that makes mixing work, so S4 hides it. Similar-looking groups: Fold, Nets, Slice, Views and Hidden; Rotate, Spin and Orient; Knots with Nets and Fold; Homing and Reckon; each test format with the exercises it resembles, except Angles.

Free play

Choose the exercise, the variant, fixed settings or adaptive items, and a timer on or off. Everything is logged; no rating changes.

Path

Ten stages per exercise, one per tenth of its difficulty range, practised in blocks. A new stage opens with its Watch demo and names its settings. A stage passes when all three hold: at least 10 items at the stage; a moving average of right answers (each new answer weighs 15 %) reaches 80 % above chance, which is 0.80 for open answers, 0.85 for four options and 0.90 for yes/no; and your rating predicts at least 80 % above chance on the stage. Simple moving-average rules like this nearly match model-based mastery rules (Pelánek & Řihák, 2018).

Passed stages come back for a 5-item recheck after 3, 7, 14, 30 and 60 days, then every 90; a stage holds while your rating predicts at least 70 % above chance on it, or reopens. No text ever calls a skill finished.

Speed

Opens for an exercise once a Path stage is passed, or after 60 answers with a settled rating. Items come from settings you already answer at least 85 % right above chance, because the deadline is the challenge; it starts at 1.5 times your median time and adapts. Speed keeps its own timed rating. Its summary gives right answers in time per minute and a balanced integration score, which resists speed–accuracy trade-offs (Liesefeld & Janczyk, 2019).

Threshold

One setting moves on a staircase while the rest stay fixed. Results keep every trial's exact value and never change a rating. Each run starts a little easier than your last estimate.

- **Track Standard:** a staircase after Legault, Allard and Faubert (2013), with 8 spheres, 4 targets and 8 s of tracking as in Faubert (2013); a 2 s cue, a still view, 20 trials. The speed goes up after a fully right trial and down after any miss, in steps of 0.05 $\log_{10}$ (about 12 %). The threshold is the geometric mean of the speeds at the last 4 reversals: the speed you track fully right about half the time. Speeds are in Parallax's own unit, so they are not comparable with other trainers.
- **Span length:** two right answers lengthen the sequence and one miss shortens it, which settles near 70.7 % right (Levitt, 1971).
- **Spin deadline:** a staircase on the time allowed.
- **At your own target:** a weighted up-down staircase (Kaernbach, 1991) settles at your style's success rate.

Which mode, when

IF YOU WANT	USE
a steady routine with mixed practice	**Train**, in the Balanced style
clear steps, one stage at a time	**Path**
a number to compare over months, from a research-style staircase	**Threshold**: Track Standard, Span length
to get faster at what you already do well	**Speed**
to explore one setting, or play what a friend played	**Free play**, or a challenge code
to rehearse the six test formats against one clock	**Section practice** (page 28)
to know whether your practice carries over	the **Spatial Check**, every two weeks

WHAT THE SIMULATIONS SHOW

200 simulated learners with their own abilities, learning curves, lapses and breaks practised for 60 days in each mode (run again for 1.1 on 13 November 2026). Success on target items stayed within 2.2 points of the target in every style. The 95 % ranges held the true ability 95.8 to 96.3 % of the time in Train, 98.6 % in Path, and 92.2 % with deliberately wrong item difficulties. In Path, 95.7 % of learner and exercise pairs passed stage 1 and 4.7 % reached stage 10. Track Standard thresholds had a typical error of 0.12 $\log_{10}$, Span lengths were within one block 73 % of the time, and one Spin deadline run is a rough guide. These are simulations of the engine, not studies of people: they show that it does what it says, not that it helps.

The active-time clock

Since 1.1 one clock measures every session, and the same minutes appear everywhere: in the HUD, the summary, the Log, the share card, Progress and the weekly goal.

What it counts, and what stops it

It starts with a session's first item and runs during items, feedback, show-me and tutorials. It stops:

- while the pause sheet is open;
- while any other sheet or dialog is open, such as How to play or a strategy card;
- while the tab is hidden, or the phone's screen is off;
- after 60 seconds without input. Those 60 seconds count, nothing after them does, and the next touch, click, key or mouse movement starts it again.

How a session ends

Train, Path and Speed sessions, and Free play without a set length, last their length in active minutes. When about one item is left at your own pace, an easier closing item comes; an item that runs past the end closes the session right after it. A quick player gets more items, not a shorter session: the plan grows when its items run out with time left. The HUD reads "about 12 min left" (rounded down), "less than 1 min left" or "last item", refreshed every 5 seconds outside the stimulus. Speed plans its items from their deadlines.

One way of writing minutes

Whole minutes, rounded: "under 1 min", "7 min", "1 h 05 min", the same on every screen. Progress shows practice this week and in all. A Spatial Check is measurement, not practice: its minutes stay out of every practice total.

Reloads

A reload or a closed tab ends the session with what was played. Reopening the app shows that session's summary, for up to 12 hours, instead of starting a new session by itself.

The Daily Challenge

The day's scored attempt counts only when all 8 items are answered. One left part-way, by "End now", a reload or a closed tab, goes on where it stopped when you open it again that day, and unanswered items are never counted wrong. The day is your local date, the same boundary everywhere in the app.

The weekly goal

A session counts towards it with 10 answers or 5 minutes of active practice. A Spatial Check is a measurement and never counts.

> HOW IT WAS CHECKED
>
> A browser test plays five-minute Train, Path and Speed sessions with quick and slow stretches, a 100-second pause, a 100-second hidden tab and a 70-second idle stretch. Each session ended within one item of five active minutes, and the summary, the share card, the Log, Progress and Today all showed "5 min".

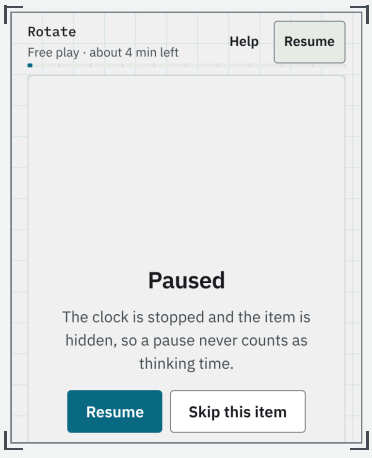


FIG. 4.1 Paused during a five-minute Free play on a phone: the clock stops and the item is hidden.

Section practice

A rehearsal of the test conditions for the six DAT-style formats (page 21): the parts in the order of the Perceptual Ability Test in the Dental Admission Test (DAT®), one clock, no feedback until the end. It practises pacing and switching between formats; the exercises themselves teach each format.

SIZE	ITEMS	TIME
Mini	5 a part, 30 in all	20 minutes
Full	15 a part, 90 in all	60 minutes

Both keep about 40 seconds an item, the real test's average pace. A part that cannot run on this device is left out, and the section keeps the same pace for the rest.

Forms

Each size has numbered forms. A form holds the same items for everyone with the same version of the exercises, so friends can compare one. Its items sit in the middle of each exercise's difficulty range, 35 to 65 % of the way up. The setup offers the next form as new to you, and marks the formats you have never practised, with a way to their Watch, Try, Go.

During a section

- Each part opens with its rules; the clock waits while you read them.
- Choosing an answer moves on. Back, Flag and Skip sit under the item, and Items shows the part's items, the answered ones marked.
- You can go back within the part you are in. Moving on to the next part closes this one for good; first a card lists what is unanswered or flagged.
- The clock stops for Pause, any open sheet, a hidden tab and five idle minutes, and warns at 5 and at 1 minute left.
- A voided item (a stutter, the phone turned) does not count, and a replacement joins the end of its part.
- A reload offers to resume the run where it stopped.

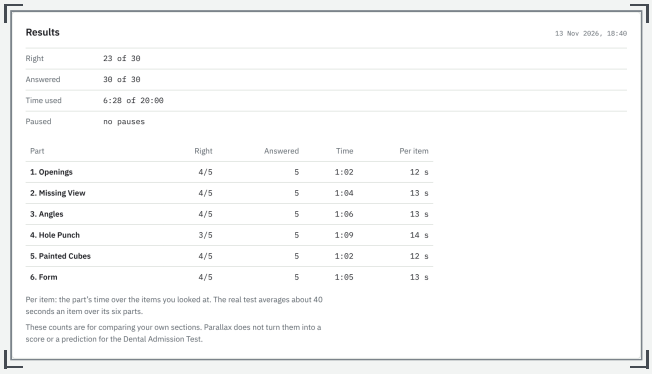


FIG. 4.2 The review after a mini section, played by a simulated player: results and pacing per part.

The review

After the last part, when the time runs out, or when you end the section from the pause sheet: right and answered out of all items, the time used and any pauses, and for each part the right answers, the time and the time per item. Then the items to look at again, each replayed with your answer and the right one marked, and its show-me.

> NO SCORE, NO RATING
>
> Section practice never changes a level. Its counts are for comparing your own sections; Parallax turns them into no score, scale value or prediction for the test. Its trials are logged as Section practice, and a full section with every part reached and at least half of its items answered earns the achievement "Full section".

DAT® and Dental Admission Test are marks of the American Dental Association. Parallax is not affiliated with or endorsed by the ADA. Every item is generated by the app; none is taken from the DAT.

CHAPTER 05

Transfer

Transfer means getting better at something you did not practise. It is the question that matters most and the one with the weakest answers, so Parallax measures it for you instead of promising it.

Near, mid and far

Transfer is not one thing. Barnett and Ceci (2002) proposed judging any transfer claim along several dimensions of content and context, each running from near to far. Parallax keeps practice apart from three rungs of transfer, and says which one each claim is on:

PRACTICE	The exercises you train, with new items each time. Expected to rise most, and not transfer.	shown by your levels
NEAR	The same kind of task in another form, such as a published paper-folding test after Fold.	not measured here
MID	Spatial tasks you never train: other formats, other objects, other kinds of spatial thinking.	measured by the Spatial Check
FAR	School, work, sport, driving, everyday life, intelligence.	cited from research; never promised

What research shows

Spatial tests: real gains, with caveats

The largest meta-analysis of spatial training pooled 206 studies (Uttal et al., 2013). The average effect was g = 0.47. Untrained spatial tests gained as much as the trained task (0.51 within the same kind of task, 0.55 across kinds, 0.45 on the trained task itself), and delayed tests (0.44) held up about as well as immediate ones (0.48).

The caveats matter as much. Control groups gained g = 0.45 on their own, from retesting and filler tasks. Against controls who did other spatial tasks, the training effect was 0.33. Published studies showed larger effects than unpublished ones (0.56 against 0.39). Only 4 of the 206 studies tested later than a month.

Games and general ability: little or none

Action video games looked good for spatial skills in 2018 (g = 0.45; Bediou et al., 2018); in the 2023 update the estimate was 0.26, with an interval from −0.27 to 0.79 (Bediou et al., 2023). Across game-training studies the effect was g = 0.07, and about zero after correcting for unpublished results (Sala et al., 2018). Cognitive training in general shows minimal effects on general cognition (Sala & Gobet, 2019).

Mathematics and science: small

Spatial training improved mathematics by g = 0.28 (29 studies, 3,765 people aged 3 to 20), mostly on maths close to the training and more with hands-on materials (Hawes et al., 2022). In the one randomised college trial (77 students, 12 hours), the physics grade rose slightly (d = 0.32), and no differences remained eight months later (Miller & Halpern, 2013). Causal evidence for effects on STEM grades, retention or degrees is still missing (Stieff & Uttal, 2015).

Familiarity and expectations

Retaking a test raises scores by 0.33 standard deviations at the second sitting and 0.50 by the third, and then the gain stops; with alternate forms it is smaller (0.23 against 0.37 for identical forms; Scharfen et al., 2018). Expectations matter too: after the same hour of training, only people recruited with a suggestive advertisement improved on reasoning tests (Foroughi et al., 2016).

IN ONE SENTENCE

Practising spatial tasks reliably improves spatial test performance, including on untrained tests, for at least several weeks; gains in mathematics are small and shown mainly in children and teenagers, gains in STEM grades are small and not firmly established, and there is no good evidence of higher general intelligence. So Parallax measures your transfer instead of promising it.

How Parallax trains for transfer

FEATURE	WHAT IT DOES	WHY
Breadth	Exercises across visualisation, speeded rotation, closure, visual memory, perspective taking and navigation, in three of the four kinds of spatial task.	Transfer across kinds of task was as large as within them (Uttal et al., 2013).
Variability	Variants, settings, viewpoints and objects change from item to item; no seed repeats.	Rotation and paper-folding practice transferred to new items and to each other: the gain was in the process, not the items (Wright et al., 2008). Variety helps generalisation (Raviv et al., 2022).
Mixing without names	Interleaving stage S4 alternates similar-looking exercises without a name banner, so you choose the transformation yourself.	Interleaving helps most when categories look alike (Brunmair & Richter, 2019).
Two strategies, and when to switch	Every exercise has an imagery card and an analytic card; occasional one-tap check-ins log which you used.	Teaching both kinds closed a sex gap in organic chemistry, while either alone did not (Stieff et al., 2014); strategy switching relates to rotation scores (Nazareth et al.); elsewhere this is untested, so the cards say to try both.
Feedback that explains	After a miss, show me replays the solution from the grading model, slowly, with pause and step; tags say why each wrong option is wrong.	Feedback with task and process information had d = 0.99, against 0.46 for right/wrong plus the answer (Wisniewski et al., 2020); animation beat static pictures (d = 0.37; Höffler & Leutner, 2007); seeing the informative views mattered more than interacting (Keehner et al., 2008).
Bridges	Each exercise names where its skill is used, with the limitation beside it. These notes are context, not a tested route to transfer.	Making connections explicit ("bridging") was proposed as a way to help transfer (Perkins & Salomon, 1988); it is untested for spatial training.
Error feedback without fixes Reckon	No landmark corrects the walk; after each answer the angle and distance errors, the drift per metre and your habit over the last 20 routes.	In a lab version of the task, feedback after every trial removed the overshoot of distance within one session (Noel et al., 2020); whether that lasts or transfers is untested.
Hands on	Off-screen challenges for Fold, Nets and Views, for variety.	In studies of children, hands-on training went with larger maths gains (Hawes et al., 2022; Gilligan-Lee et al., 2023); these challenges are untested, and Parallax makes no maths claim.
Dose	20-minute sessions, three a week, by default.	Several effective programmes ran about 10 to 20 hours, and more hours have not been shown to help more (Hawes et al., 2022); more than three sessions a week added nothing in older adults (Lampit et al., 2014).

One honest gap: no study isolates animated against static corrective feedback for mental rotation, so show-me is a design bet. Parallax does not yet log each use, so this stays open.

How the Spatial Check measures transfer

Probes you never train

The Spatial Check is a battery of short tests, the probes, whose formats no exercise uses. They never give feedback, and they are never used as warm-ups. Three spatial probes are one step away from training (mid transfer), two test kinds of spatial thinking that no exercise trains, and a word task serves as the control.

A short adaptive test

Each probe has 16 items and takes 3 to 5 minutes. The first two items are easy; each later one aims at about 60 % above chance under the current estimate: neither trivial nor hopeless. Your score θ and its standard error come from the exact posterior with the same starting assumption every time, and forms A, B and C (the same blueprint, different seeds) rotate, so checks stay comparable. Accuracy and the median time of right answers are reported separately. Each item has a calm time bar; a time-out counts as wrong.

When to take it

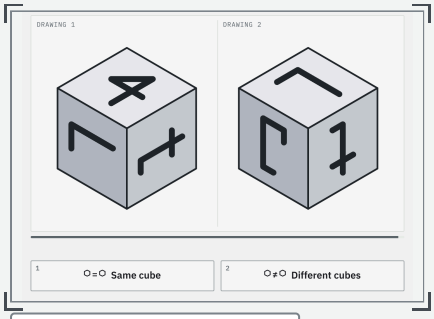
- **Two baselines**, the second at least about 12 hours after the first; a third when the two differ by more than two combined standard errors.
- **Then every 14 days or 8 sessions**, whichever comes later.
- **A delayed check** about four weeks after training stops or slows, where the research is thinnest.

Before the first baseline, Parallax asks once what you expect from training. The answer is only logged, because expectations alone can raise scores (Foroughi et al., 2016).

The control

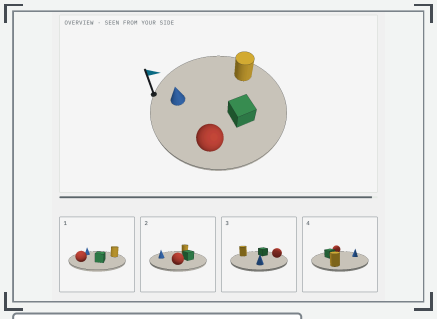
Word Links (PR-0) is about words, not space. Training should not change it, so its change estimates what comes from getting used to being tested and from motivation.

The six probes



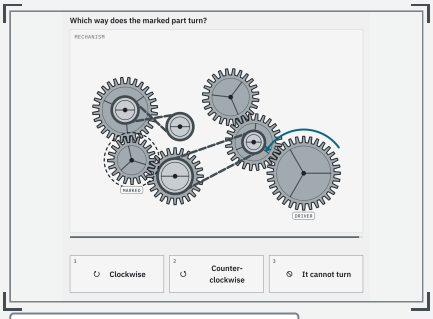
PR-1 · Cube Match · Vz

Could the second drawing be the first cube, turned? Each face has its own symbol. Mid transfer: Nets trains unmarked faces, Rotate trains rotation.



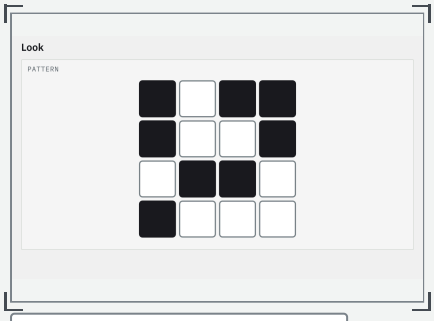
PR-2 · View Match · PT

Which picture shows the table from the flag? Mid: Orient trains pointing on a map, Views trains flat line drawings.



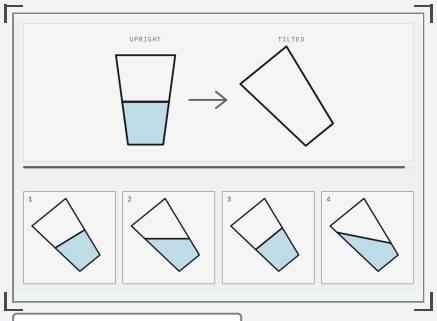
PR-3 · Mechanisms · MA

Gears, belts and pulleys: which way does the marked part turn, does the weight rise, how fast? Mental animation, which no exercise trains.



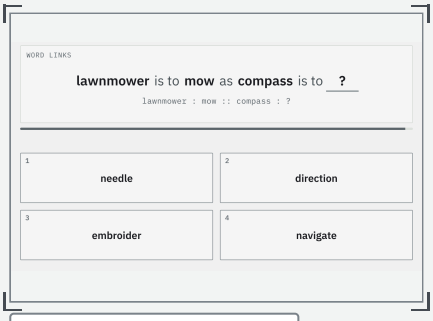
PR-4 · Pattern Memory · MV

Look at a pattern for 3 seconds, then mark the squares that were filled. Mid: Span trains sequences, not patterns.



PR-5 · Level · Vz

Which picture shows the water surface, or a hanging string, the way it would really be? Extrinsic-static, a kind of task no exercise trains. Optional.



PR-0 · Word Links · VB

A is to B as C is to ...? Original verbal analogies: the control.

The four spatial probes PR-1 to PR-4 also form a **composite**, their mean, which is more precise than any one of them. Level (PR-5) is offered as an extra. Every Mechanisms item needs more than the rule that neighbouring gears turn opposite ways, which people learn to use instead of picturing the motion (Hegarty, 2004): belts, shared axles, side branches or jams. Every item of every form is generated and solver-checked like the training items.

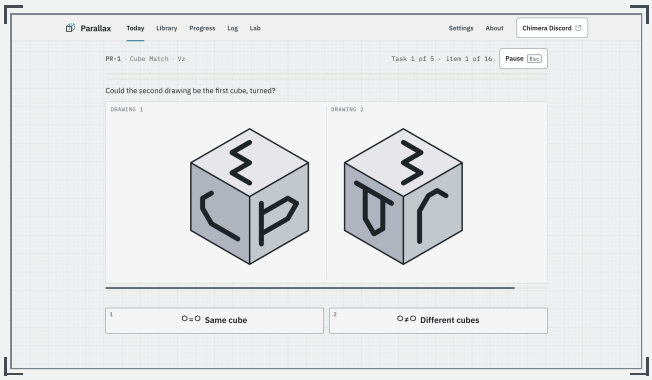


FIG. 5.1 A probe in the Spatial Check: instructions and one example first, then items without feedback.

Taking a check well

- Find 15 to 25 quiet minutes. A pause stops the clock; an item paused while it plays is replaced. One skip per task is free; after that a skip counts as wrong.
- Later checks first ask whether you have practised or taken other tests with such puzzles, or with word analogies, since the last one: logged, never scored.
- Answer as well as you can, and guess when you are unsure. The items adapt, so expect roughly 6 to 8 in 10 right: the difficulty you reach carries the information, not the count.
- Do not look up or practise the probes. Their value is that they are new.
- Expect "unclear yet" for a while. That is the honest answer for one person and a few checks.

Reading the report

For each probe and for the composite, Progress shows your **change** since the average of your baselines, with a 95 % interval; accuracy and time separately; the **control's change**; the **familiarity estimate**; and, beside them, the level changes on the exercises you trained: practice gains, not transfer. It never shows an IQ, an index, a percentile or a brain age.

The familiarity estimate

Retest gains roughly halve after the second test and then stop (Scharfen et al., 2018). So your own gain from the first to the second baseline, called *G*, predicts what familiarity alone will still add: about another half of *G*. Change is measured from the average of the two baselines, which already sits *G*/2 above the first, so the familiarity part of any later change is *G*. Two baselines are what make this estimate possible. The estimate is itself measured from two short tests, so its uncertainty is counted too.

Under the chart, a table ("Why these words") repeats each probe's numbers.

Three words, with exact rules

`beyond familiarity` when both hold: the lower end of the 95 % range of your change minus the familiarity estimate is above zero, counting the uncertainty of both; and the lower end of your change minus the control's change is above zero. Practice is then one likely reason; one person's checks cannot rule out expectations or chance.

`no change` when even the upper end of your change beyond familiarity is below 0.25 points on the probe's scale.

`unclear yet` otherwise: at least one comparison (with familiarity or with the word task) still includes zero. This is the usual result and not the same as no progress; a change shows once it is larger than this range.

In the simulations run for 1.1, none of the 1,600 learners whose probe ability did not change got "beyond familiarity" (in 1.0, 1 in 1,600). Of learners whose probe ability did rise (by half their training gain), 96.5 % were still "unclear yet" after 60 days. Each check is compared with the baselines on its own, so taking more checks does not narrow the range; the composite is more precise than any one probe.

ONE PERSON, OR MANY

Each check is short, so most people will see "unclear yet" for weeks, often for months, even when practice is helping. That is the honest precision of a check of 15 to 25 minutes, not a fault. The place where transfer can really be measured is pooled data: many people's checks together, each with its familiarity estimate and its control task, can show what one record cannot. If you would like to help, export your trials from the Lab (Research export) and share the file in the Chimera Discord, discord.gg/chmr. Nothing is ever sent automatically: you choose what to share.

A worked example

ILLUSTRATIVE NUMBERS, INVENTED FOR THIS GUIDE

SCORES (± STANDARD ERROR)	BASELINE 1	BASELINE 2	WEEK 4
Spatial composite	−0.30 ± 0.20	0.00 ± 0.20	0.75 ± 0.20
Word Links (control)	0.10 ± 0.25	0.30 ± 0.25	0.40 ± 0.25

1. Baseline average: (−0.30 + 0.00) / 2 = −0.15, standard error $\sqrt{(0.20^2 + 0.20^2)} / 2 = 0.14$.
2. Change: 0.75 − (−0.15) = **0.90**, standard error $\sqrt{(0.20^2 + 0.14^2)} = 0.24$, so the 95 % interval is 0.42 to 1.38.
3. Familiarity: $G = 0.00 - (-0.30) = 0.30$, itself measured with standard error $\sqrt{(0.20^2 + 0.20^2)} = 0.28$. Beyond familiarity: 0.90 − 0.30 = 0.60, standard error $\sqrt{(0.24^2 + 0.28^2)} = 0.37$, interval −0.13 to 1.33. Lower end below zero: **not yet**.
4. Control change: 0.40 − 0.20 = 0.20, standard error 0.31. Spatial change minus control change: 0.90 − 0.20 = 0.70, standard error $\sqrt{(0.24^2 + 0.31^2)} = 0.39$, interval −0.07 to 1.47. Lower end below zero: **not yet**.
5. Verdict: `unclear yet`. Both comparisons still include zero: a gain this size could still be familiarity, or getting used to being tested.
6. Had the check come out at 0.95 instead, with the control unchanged, the change would be 1.10, with the same standard errors (each check is compared with the baselines on its own): 0.07 to 1.53 beyond familiarity and 0.13 to 1.67 above the control. Both lower ends above zero: `beyond familiarity`.

The report in the app

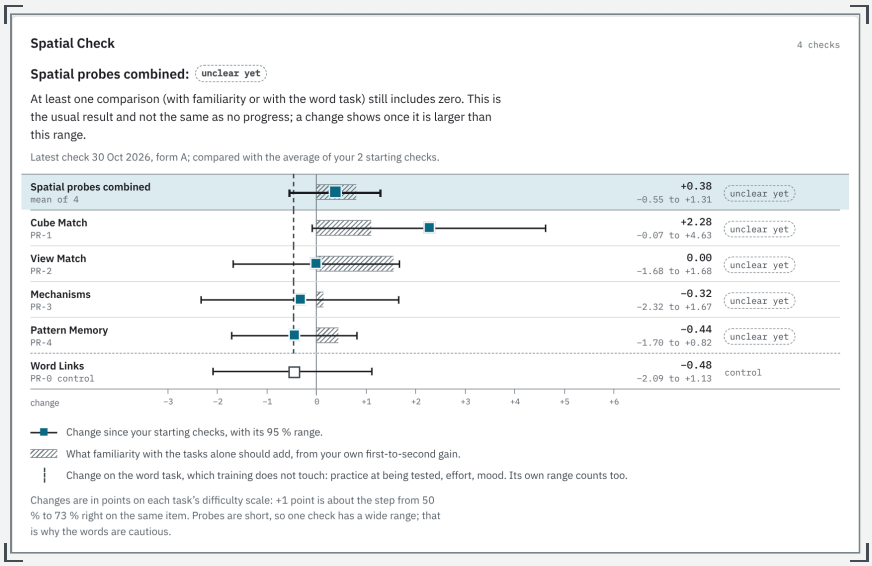


FIG. 5.2 The Spatial Check report in Progress, after six weeks of simulated practice: every probe and the composite still read "unclear yet".

Reading the chart

- Rows: the composite on top, then each probe.
- The marker and its line are your change since the baseline average, with its 95 % range.
- The hatched band is what familiarity alone should add, from your own first-to-second gain.
- The dashed line is the change on the word task, which training does not touch.
- Changes are in points on each task's difficulty scale: +1 point is about the step from 50 % to 73 % right on the same item.

The rest of Progress

Levels for every exercise with their ranges, the real settings and the Path stage, also grouped by ability and by kind of task. **Learning curves** with a noise band, one dot per practice day. **Personal bests**: your highest level after a whole block of 10 whose success reached the level reference. The **weekly goal** calendar and streak, the **achievements** earned and still open, and the backup line. Notes beside the report flag a check taken on another kind of screen than your starting checks, and Level once you have practised Angles since them.

CHAPTER 06

Motivation

Parallax wants you to come back because practice feels worthwhile, not because a counter nags you. On a few points, the research is clear enough to guide that.

What the research says

Autonomy, competence, relatedness

Self-determination theory holds that people enjoy and return to activities that meet three needs: choosing for themselves, feeling able, and feeling connected. In four studies of video games, felt autonomy and competence predicted enjoyment and wanting to play again; only competence predicted actually choosing to keep playing, and in an online game relatedness added to both (Ryan, Rigby & Przybylski, 2006).

Informing, not controlling

Across 128 experiments, positive feedback that informs raised free-choice persistence (d = 0.33) and interest (d = 0.31). Expected tangible rewards lowered free-choice behaviour (d = −0.36), rewards for simply taking part most of all (−0.40) (Deci, Koestner & Ryan, 1999).

Feedback about the task, not the person

Feedback helped performance on average (d = 0.41) but made it worse in over a third of cases, typically when it pulled attention from the task to the self, as comparisons with others do (Kluger & DeNisi, 1996).

Game elements help, modestly

In a meta-analysis, gamification improved learning outcomes (g = 0.49), motivation (0.36) and behaviour (0.25); in the methodologically rigorous studies only the learning effect held (Sailer & Homner, 2020). Learners liked performance feedback, recognition and help with goals, and disliked anxiety and jealousy (Bai, Hew & Huang, 2020).

Streaks cut both ways

An intact streak raises later engagement compared with a broken one, even when behaviour was identical; the break hurts more when people blame themselves and less when the streak can be repaired (Silverman & Barasch, 2023). Goals pull effort as they come closer, and visits often end right after a reward (Kivetz, Urminsky & Zheng, 2006), so a session that ends on its summary is a natural place to stop.

What Parallax includes

ELEMENT	WHY
Honest levels with the real settings and a likely range	Competence feedback that informs (Ryan et al., 2006; Deci et al., 1999).
Learning curves with a noise band; personal bests: the highest level after a strong whole block of 10	In one experiment, badges, leaderboards and performance graphs together raised felt competence (Sailer et al., 2017); Parallax keeps the graphs and drops the leaderboard. Whole blocks keep a best from being luck.
A weekly goal you set (3 by default, 2 to 6), on any days, with a calendar of days practised	Your own goal; more than three sessions a week added nothing in older adults (Lampit et al., 2014).
A weekly streak with automatic grace (one missed week in four) and free repair (goal + 1 sessions the next week); no daily streak	Breaks hurt less when they can be repaired (Silverman & Barasch, 2023).
21 achievements for skills and good practice (all on the next page)	Finite and visible criteria, tied to skill or to practice that helps measurement. None says "mastered".
Session summaries against your own past, with an honest line about noise and one suggested next step	Self-referenced, task-focused feedback (Kluger & DeNisi, 1996).
Every exercise open from day one; five ways to practise; three styles	Choice: autonomy (Ryan et al., 2006).
The Daily Challenge, challenge codes for friends, share cards carrying the Discord link	Relatedness without ranking; competition combined with collaboration beat competition alone (Sailer & Homner, 2020).

What Parallax avoids

LEFT OUT	WHY
Points, coins or rewards for time spent in the app	Rewards for taking part lowered free-choice motivation (d = −0.40; Deci et al., 1999).
Global leaderboards and rankings	Comparison pulls attention to the self (Kluger & DeNisi, 1996); learners report jealousy and anxiety (Bai et al., 2020).
Daily streaks, loss-framed warnings, countdowns, late-night reminders	Broken streaks lower engagement, most when people blame themselves (Silverman & Barasch, 2023).
Brain age, IQ, percentiles, fake head starts	They would claim more than the evidence supports, and percentiles of self-selected users mislead.
Exercises locked behind grinding	Grinding to unlock is controlling. Every exercise is open from day one; only Path stages and Speed open, by performance.
Autoplay of the next session; "are you sure you want to quit?"	UK guidance for services children use asks for neutral "keep playing" choices and pausing without losing progress (ICO, Age appropriate design code).
Celebrations on every right answer	A brief tick and the time; celebrations are kept for milestones such as a new level or a personal best.

The weekly goal, exactly

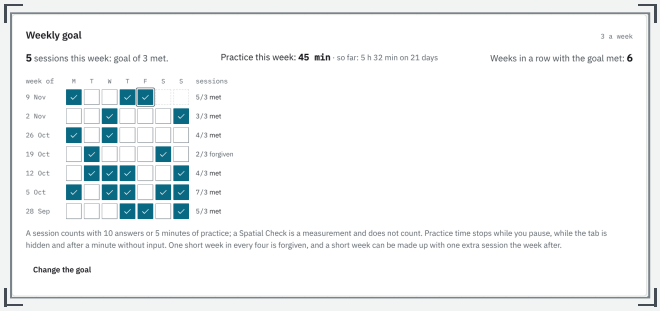


FIG. 6.1 The weekly goal in Progress: days practised and weeks met.

Weeks run Monday to Sunday. A session counts when it has 10 answers or 5 minutes of active practice; a Spatial Check never counts. One missed week in any four is forgiven automatically, and a missed week that grace cannot cover is repaired for free by doing one session more than your goal the following week. It is recomputed from your sessions every time, so it cannot drift, and it never warns about loss.

Achievements, exactly

Achievements reward skill and good practice, never time: only right answers count for skill milestones, and stuttered or skipped trials never count. **Good practice:** every exercise tried; the weekly goal met, once and in four weeks; 12 practice days within 90; two baselines; three checks at least 12 days apart; a check three weeks after training slowed; both strategies reported on one exercise; a backup downloaded; a complete Threshold run; a full Section practice. **Skill:** Path stages 3 and 10; a stage still holding at its 30-day recheck; level 50 once settled; level 20 on every exercise; a right answer from the hardest tenth of a range; five targets among ten spheres tracked fully right; a span of seven; a full-length Speed session; a new best block. Both "every exercise" milestones leave out the six opt-in test formats.

Feedback after an answer

- **Within half a second:** an icon, a word and a colour, never colour alone; the right answer marked; tags that say what each wrong option shows.
- **After a miss:** a slow show-me, built from the model that graded the answer, with pause, step, replay and close. After three automatic replays on an exercise, it waits for **Why?**, since guidance that helps beginners can hinder experienced learners (Kalyuga et al., 2003).
- **After a slow right answer:** See solution.
- **After two misses on similar items:** a strategy card is offered, never forced.
- **Wording** is about the task ("the short arm ends up on the left"), never about you, and never compares you with others.

RELATEDNESS, THE CHIMERA WAY

Parallax has no social network of its own. The community lives on Discord: share a Daily Challenge line or a card, swap challenge codes, compare notes on strategies, suggest exercises. discord.gg/chmr

CHAPTER 07

The research in one table

Each row: what research found, its numbers, and what Parallax does about it. Full references are in chapter 13.

FINDING	NUMBERS	DESIGN RESPONSE	SOURCE
TRAINING AND TRANSFER			
Spatial training works and transfers to untrained spatial tests	g = 0.47 over 206 studies; transfer 0.51–0.55; delayed tests 0.44	Train broadly; measure on probes never trained	Uttal et al. 2013
Controls gain too; strong controls shrink the effect	controls g = 0.45; against spatial-task controls 0.33; published 0.56, unpublished 0.39	Judge change against a control probe and a retest estimate, never against zero	Uttal et al. 2013
Retesting raises scores by itself	+0.33 SD at test 2, +0.50 by test 3, then flat; alternate forms 0.23 vs identical 0.37	Two baselines; alternate forms A, B, C; the familiarity estimate	Scharfen et al. 2018
Transfer to mathematics is small	g = 0.28 (29 studies, N = 3,765); larger near the training and with manipulatives	Far transfer cited, never promised; no maths claim	Hawes et al. 2022
STEM effects are small and can fade	physics grade d = 0.32; no differences at 8 months; causal STEM evidence missing	No claims about grades, work or careers	Miller & Halpern 2013; Stieff & Uttal 2015
Action games: not robust	spatial g = 0.45 (2018), 0.26 with interval −0.27 to 0.79 (2023)	No game claims	Bediou et al. 2018, 2023
No gain in general ability	game training g = 0.07, about 0 after bias correction; minimal effects on general cognition	No IQ, no index, no brain age	Sala et al. 2018; Sala & Gobet 2019
Expectations raise scores	only the suggestively recruited group gained (d = 0.50 and 0.39)	Neutral wording; expectations logged at baseline	Foroughi et al. 2016
Practice that varies is process-based	rotation and folding practice transferred to each other and to new items	Variants, varied settings, mixing without names	Wright et al. 2008
Strategies matter	imagery plus analytic training closed a sex gap; strategy switching r = .50 with rotation scores	Two strategy cards with when to switch; one-tap check-ins	Stieff et al. 2014; Nazareth et al.
Hands-on materials help maths transfer in children	manipulatives beat none; physical gave larger, more consistent maths gains	Off-screen challenges for Fold, Nets, Views, for variety only; untested, no maths claim	Hawes et al. 2022; Gilligan-Lee et al. 2023
FEEDBACK AND TUTORIALS			
Feedback that explains works best	high-information d = 0.99; right/wrong plus answer 0.46; praise or rewards 0.24	Show me after errors; tags on every option	Wisniewski et al. 2020
Animation and the right views	animation vs static d = 0.37; seeing informative views mattered more than interacting	Slow, pausable show-me from the grading model	Höffler & Leutner 2007; Keehner et al. 2008
Guidance fades with expertise	help for novices can hinder experienced learners	Automatic replays fade to a Why? button	Kalyuga et al. 2003
Tutorials: just in time	in a complex game, context tips +40 % levels over a manual; forced steps no benefit (over 45,000 players)	Watch, Try, Go; short; always skippable	Andersen et al. 2012

FINDING	NUMBERS	DESIGN RESPONSE	SOURCE
DIFFICULTY, SCHEDULING AND DOSE			
A success target, not a law	85 % optimal for two-choice gradient learners (theory); 60–70 % rated right by learners; more success, more volume	$p^* = g + (1 - g) \times k$; Comfortable, Balanced, Stretch	Wilson et al. 2019; Pelánek et al. 2017; Jansen et al. 2013
Difficulty can be predicted from item features	six features reproduced rotation difficulty at r = .90	Item difficulty from settings; trials log predictions for a refit	Rajeb et al. 2025; Fischer 1973
Ratings must keep learning	fixed-threshold methods lag a learner; tracking methods follow drift	A Bayesian rating with process noise per answer and per idle day	Doll et al. 2015; Pelánek 2016
Interleaving helps, most for similar categories	g = 0.42 for inductive learning; grows with similarity	Five stages; similar-looking exercises mixed without names	Brunmair & Richter 2019
Raise interference gradually	blocked → serial → random beat both pure schedules	Stages S1 to S5, with a step back when switches hurt	Porter & Magill 2010
Mixed practice feels worse but lasts	89 % vs 60 % during practice; 20 % vs 63 % a week later	Progress shown in levels and checks, not session accuracy	Rohrer & Taylor 2007
Learning-progress bandits need limits	reached higher levels than an expert sequence on three of four exercise types but starved the fourth	Round-robin weighted by progress, 20–50 % shares, 15 % random	Clement et al. 2015
Simple mastery rules are enough	moving averages nearly match model-based criteria	Path's gate: moving average plus rating	Pelánek & Řihák 2018
Speed–accuracy scores	BIS resisted trade-offs; IES and RCS did not	Speed reports BIS against your first timed sessions	Liesefeld & Janczyk 2019
Dose and frequency	several effective programmes ran about 10–20 h; more hours not shown to help more; older adults: more than 3 sessions a week no extra benefit	20 minutes, 3 sessions a week by default (2 to 6), all logged	Hawes et al. 2022; Lampit et al. 2014
TASKS AND TEST HYGIENE			
Paper-folding shortcuts	about 22 % mainly unfold; most count holes per fold	Same hole count on every option; shortcut solvers near chance	Burte et al. 2018
Cross-section errors	the egocentric shape is the most chosen error (.23); oblique cuts harder	An egocentric option in every item	Cohen & Hegarty 2007
Rotation strategies	global shape predicts accuracy; counting cubes predicts errors (r = −.48)	Mirror options with identical cubes; strategy cards	Hegarty 2018
Wechsler retest	perceptual-reasoning subtests show the largest practice effects	No Block Design or Visual Puzzles clones	Canivez 2010
3D-MOT: large task gains, weak transfer	trained task d = 1.6–2.0; 29 studies, none preregistered; 2 of 3 transfer tests (sport, driving) null; no transfer to soccer play	Track labelled attention practice; the research protocol in Threshold	Harris et al. 2020; Vater et al. 2021; Romeas et al. 2025
Tracking is not a CHC visual-processing (Gv) ability	loads with attention, visual working memory and span	Track kept outside the spatial abilities	Treviño et al. 2021
MOTIVATION			
Game elements help modestly	learning g = 0.49; motivation 0.36, behaviour 0.25 (not significant in rigorous studies)	Only informational elements	Sailer & Homner 2020
Rewards for taking part backfire	expected tangible rewards d = −0.36; engagement-contingent −0.40	No points for time	Deci et al. 1999
Comparison can hurt	feedback worsened performance in over a third of cases	No leaderboards; self-referenced summaries	Kluger & DeNisi 1996
Broken streaks demotivate	lower later engagement; less when repairable	Weekly streak with grace and free repair; no daily streak	Silverman & Barasch 2023

FINDING	NUMBERS	DESIGN RESPONSE	SOURCE
NAVIGATION WITHOUT LANDMARKS (1.1)			
Feedback can remove dead-reckoning bias, at least within a session	with end-point feedback, distance gain 1.10 → 0.97, angular gain 1.49 → 1.09; practice and feedback not separated	Reckon: angle, distance, drift and habit feedback after every answer; tolerances tighten with skill	Noel et al. 2020
Navigation training transfers little	42 sessions: navigation +1.42 SD (young adults); other tests at most a non-significant trend (+0.29 SD, p = .072)	Homing and Reckon experimental; no wayfinding claim	Lövdén et al. 2012
The body's own senses help	in free-walking VR, adding walking to vision halved the position error (22.5 → 10.6 m)	"walking adds body senses that a screen cannot give"	Embodied Navigation (preprint) 2025
Without body cues, sides get swapped	37.5 % of naive people consistently swapped left and right	a named hint for swapped sides; tutorial items for one turn and one distance	Riecke & Wiener 2007
A few trials give an unreliable score	6 trials: test–retest r < .5 for end-point and angle error	levels shown with ranges, which narrow only over many trials	McLaren et al. 2022
KNOTS (1.1)			
Knot reasoning is spatial, and partly its own skill	Knot Reasoning Task: r = .54 with mental rotation, .59 with paper folding; together about 40 % of its variance	Knots as a visualisation exercise beside Rotate and Fold	Bennett-Pierre et al. 2025
People see a knot's shape, not its strength	strength judgements at or below chance (36.9–49.6 %); naming the knot 92.5 % (chance 25 %)	only questions of shape: does a knot remain, are loops linked	Croom & Firestone 2024
Spatial ability relates to learning to tie knots	mental rotation and surgical knots tied: r = .42–.45	the "where it is used" line; no claim about tying	Brandt & Davies 2006
The unknot can be certified	no knot with a diagram of up to 24 crossings has a Jones polynomial of 1	the validator's last step; ropes of at most 12 crossings	Tuzun & Sikora (arXiv)
TEST FORMATS AND TEST HYGIENE (1.1)			
The PAT predicts dental grades modestly	corrected correlations .16–.36 with grades in 38 schools; highest for second-year technique	a limitation line beside every DAT-style text; no score prediction	ADA 2026
Coaching adds to retesting	coached .70 vs .24 SD without coaching; identical forms .46 vs alternate .24	"Also in tests" notes; take a test that matters first	Hausknecht et al. 2007
Being taught a puzzle type's rules raises scores	figural matrices: d ≥ 1.19 after a short video; d = .92 with rule information	About's Test hygiene: about one standard deviation	Schneider et al. 2020; Levacher et al. 2022
Practice gains are not gains in g	g loading against retest gain: ρ ≈ −.95 (64 studies)	gains described as staying with the kind of puzzle	te Nijenhuis et al. 2007
Spatial practice left verbal analogies unchanged	about 7 hours of rotation or folding practice: analogy errors 19.4 → 19.2 %	analogies untrained; Word Links stays the Check's control	Wright et al. 2008
Angles are judged best near 90°	thresholds 5.4° at 90°, 8.9° at 80° and 100°, 11.2° at 55° and 125°	Angles: near-right sets easiest, then sharp, then wide	Dillon et al. 2019
Classic cube-counting errors	counting only visible cubes or faces; accuracy falls with more cubes and hidden faces	Painted Cubes: the visible-face count held near chance; the tally strategy card	Kurtuluş & Yolcu 2013

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CHAPTER 08

Honest limits and test hygiene

What Parallax cannot tell you, what it deliberately avoids, and where its own design rests on judgement rather than data.

No clones of clinical tests

Parallax has no version of Wechsler Block Design or Visual Puzzles, the two subtests of the Wechsler Visual Spatial Index, and no packing or piece-assembly tasks that resemble them. On retest, the perceptual-reasoning subtests that include them show the largest practice effects (Canivez, 2010). Practising their formats would inflate exactly the scores that psychologists use, probably without changing the ability (Scharfen et al., 2018; te Nijenhuis et al., 2007), and would invite claims such as "my VSI went up". No Parallax score is called VSI, and no item, figure or layout is copied from a published test: every item is generated.

Formats you may meet elsewhere

Several exercises share a format with parts of published tests: paper folding, hidden shapes, counting cubes, cube nets, block tapping, knots pulled tight, and the six test formats of page 21, on purpose. A gain on such a test after practice here is near transfer, not mid transfer. The next page gives the formats, the size of practice effects, and what Parallax does about them.

> IF YOU TAKE A TEST
>
> Take a test that matters before you practise its kind of puzzle here, or mention your practice: for a clinical, school or job assessment with spatial tasks, say that you have practised similar tasks, so the result can be read fairly. Settings has a "Taking a test soon?" helper (page 40).

Not medical, not diagnostic

Parallax is a practice tool. It does not diagnose, treat or monitor any condition, and its numbers should not be used as evidence about one. It makes no claims about ageing, memory decline, concussion, attention disorders or any other condition. If you are worried about your memory or attention, talk to a health professional.

Hand-set item difficulty

Each exercise predicts an item's difficulty from its settings, with one step per setting level. The steps were set by hand: from published data where it exists (for Rotate, the error rates in Ganis and Kievit's validation data give about 0.016 logits per degree; Ganis & Kievit, 2015) and from judgement elsewhere. No pilot study calibrated them.

Three things limit the damage. Simulated learners whose items were deliberately mis-set (random errors of 0.5 logits plus an interaction between two settings) still got items about half a percentage point from the target, and their 95 % ranges held the true ability 91.5 % of the time. Weak-spot detection adds items where you do worse than predicted. And every trial logs its settings, the predicted chance of a right answer, the interleaving stage and the item-model version.

That last part is where the community comes in. Pooled exports (chapter 10) would let anyone refit the steps by logistic regression with a term per person, and better steps could ship in a later release, while your rating stays on your device. If you want to help, export from the Lab and share it on the Discord.

Shortcuts

Every item pool is attacked by cheap shortcut solvers: counting cubes or holes, the odd one out, the option most like the others, the shape as seen from here, gear parity, the over-under pattern of a rope. Each must score within 10 points of chance, or the exercise would train the shortcut instead of the skill. Some shortcuts are partly valid geometry; where one comes close to the bound, such items are rationed rather than banned.

A few partial strategies do better, and are kept because they are part of the skill: unfolding Fold and Hole Punch sheets by their symmetry about the creases, the format's own method, which fails where layers must be tracked; in Knots' Link items, counting the crossings between the two loops with their signs; in Painted Cubes, counting only the cubes in view, which misses the hidden ones; in Openings, checking only the face with the most features. The tests report them; the strategy cards and show-me teach the full method.

Test hygiene: formats that tests share

Practising a kind of puzzle raises scores on tests that use it. That gain is real, but it is near transfer: it says little about the ability a test is meant to measure. So Parallax says where its formats overlap with published tests, and lets you plan around a test.

How big practice effects are

- Taking the same kind of test again: +0.33 standard deviations at the second sitting and +0.50 by the third, then little more; 0.23 with alternate forms against 0.37 with identical ones (Scharfen et al., 2018).
- Coaching nearly triples it: .70 against .24 for retesting alone (Hausknecht et al., 2007).
- Being taught a puzzle type's rules: about one standard deviation on figural matrices, d = .92 to 1.19 or more (Levacher et al., 2022; Schneider et al., 2020).
- The gains stay mostly with that kind of puzzle: the more a subtest measures general ability, the smaller its retest gain (te Nijenhuis et al., 2007). They can last for years (Sommer & Arendasy, 2025).

Shared formats

EXERCISE	SHARED WITH PARTS OF PUBLISHED TESTS	IN THE APP
Fold, Hole Punch	folding and punching paper	a note
Hidden	finding a hidden shape	a note
Views (count), Painted Cubes	counting cubes in a stack	a note
Nets, Form	folding a net into a cube or solid	a note
Span	tapping blocks in order, as in clinical memory tests	a note
Openings, Missing View, Angles	openings, views and angle ranking: with Hole Punch, Painted Cubes and Form, the six DAT-style formats follow the Perceptual Ability Test on purpose	a note
Knots (Pull)	"knot or not" items in a design entrance exam, a maths competition and a research test of knot reasoning	a note
Slice, Orient, Homing, Reckon, Rotate, Spin	cross-sections, directions and maps, turned or mirrored figures	related only

Analogies. No exercise trains analogy items ("A is to B as C is to ..."). The Spatial Check's word task uses that format for a few minutes per check, without feedback. In one study, about seven hours of rotation or folding practice left verbal-analogy accuracy unchanged (Wright et al., 2008).

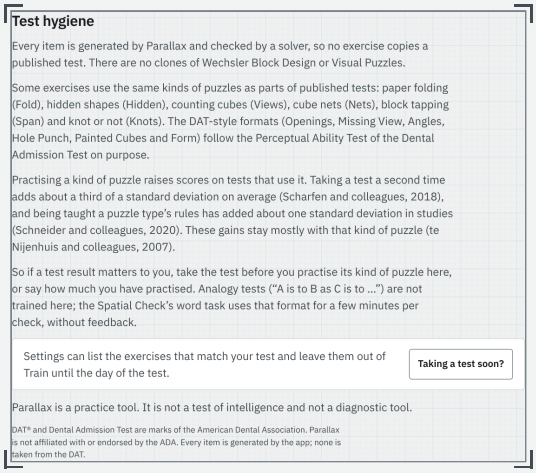


FIG. 8.1 About's Test hygiene section.

What Parallax does

- **"Also in tests"** notes on those exercises' Library pages and in How to play.
- **Later Spatial Checks** ask one question about outside tests and practice since the last check, kept beside your data and never scored: outside practice moves the Check too.
- **"Taking a test soon?"** in Settings: tick what your test contains, from paper folding to word analogies, or "not sure". Parallax lists the exercises with the same kind of puzzle, the related ones, and what it does not train. It can leave the same-kind exercises out of Train and Today's plan until the day of your test, and note the test in your record; Free play, Path, Section practice, the Library and the Daily Challenge keep them.

> IF A TEST RESULT MATTERS TO YOU
>
> Take the test before you practise its kind of puzzle here, or say how much you have practised. Compare only results taken under the same conditions. For a clinical, school or job assessment with spatial tasks, mention that you have practised similar tasks, so the result can be read fairly.

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Precision

- **One person is noisy.** Sixteen items per probe give wide ranges, and each check is compared with the baselines on its own, so taking more checks does not narrow them. "Unclear yet" is the honest and common answer; the composite of four probes is more precise than any one.
- **Levels are estimates.** Each comes with a likely range, and session-to-session noise of a few levels is normal.
- **Screens differ.** Viewing size changes how hard some tasks are, so Track, Homing, Reckon and Angles keep separate ratings for small and large screens, and every trial records its screen class. Turning a phone while an item plays voids that trial.
- **Software rendering.** On slow devices, frames can stutter. Stuttered trials are voided and never count, and after two of them quality steps down automatically.

What research has not settled

- **Dose and spacing** for spatial training: no head-to-head comparison was found, and no reliable dose-response either.
- **Session length:** weak evidence that sessions under 30 minutes did not work in older adults (Lampit et al., 2014), against efficient 8-minute sessions in perceptual learning (Molloy et al., 2012).
- **Animated against static** corrective feedback for rotation: untested.
- **Navigation training** on a screen, and **perspective-taking training** as such: no trials found. For **path integration**, no randomised training study with a transfer test was found, and phone-sized screens are untested.
- **Adults' disembedding** (Hidden): no training studies found.
- **Knot diagrams** as training: no study found.
- **Practice on the six test formats** against a control group, with the test itself as the outcome: no study found.

Parallax makes a default choice for each. Its trial log lets pooled data say more later about dose, spacing and the untested exercises; it does not yet log each use of show-me, so that question stays open.

Points left out because they were not verified

The research behind this guide was compiled from primary sources where they could be reached. About a hundred points could not be confirmed against a primary source (a paywall, a blocked page, or a search budget that ran out) and are marked as unverified in the project's notes. This guide leaves those points out or calls them "reported". Among them:

- sample sizes and follow-ups of several older training studies, including an early action-game study and a long Tetris study;
- effect sizes from a large regression-discontinuity study of an engineering spatial course;
- the strength of the links between object tracking and fluid reasoning or Corsi span;
- a chapter that frames spatial intelligence as image transformation plus strategy choice, and a study of gesture in spatial problem solving;
- the exact dimensions of Barnett and Ceci's transfer taxonomy, and mastery-learning effect sizes;
- how often adults fail the water-level task.

In the reference list, details that could not be confirmed, such as page numbers, are left out. Where a year is uncertain it is marked n.d.

ACCESSIBILITY

Everything is playable by keyboard; touch targets are at least 44 pixels; colour is never the only cue; nothing flashes faster than three times a second; reduced motion is respected outside the stimulus. Track, Homing and Reckon are moving displays by nature: Parallax warns before them, and they pause at once. Every scored item fits a 360 × 640 screen, and text inside the figures stays at 13 pixels or more on phones.

CHAPTER 09

Your data

Everything you do stays in this browser on this device. Nothing is sent anywhere: there are no accounts, no analytics and no ads. You decide what to copy, export or share.

Where it lives

Parallax keeps your record in the browser's local storage. Where storage is blocked (some private windows, some embedded previews), it runs from memory and says so on screen: progress is then lost when you close it, so back up before you leave. Two open tabs stay in step with each other. After 60 simulated days, a record took at most about 1.1 MB (typically about a third of that).

Since 1.1 the record is in format 2. A 1.0 record upgrades when 1.1 first opens it, with nothing lost; a 1.0 tab still open elsewhere then runs from memory rather than overwrite the newer record.

Back up and restore

- **Back up now** (in Settings, on Today and in Progress) saves the whole record as a file, `parallax-backup-<date>.json`. Where downloads are blocked, the same text is shown to copy.
- **Import a backup**, from the file or pasted text, replaces the progress in this browser. You can undo it once, until you close Parallax. Every import is checked and repaired against the record's schema first.
- The "last backup" line on Today and in Settings tells you how old your newest backup is.
- **Reset progress** clears levels, sessions, trials, checks and achievements but keeps your settings, and can also be undone once.

Export formats

EXPORT	WHAT IT HOLDS
Backup (JSON)	The whole record: ratings, sessions, trials, Spatial Checks with their one-tap answers, Threshold runs, Section runs, strategy check-ins, the tests you noted, the focus switches, achievements, settings. The only format that can be imported again.
Trials (CSV)	One row per trial: time, exercise and its version, mode, session, settings, seed, outcome, response time, the predicted chance of a right answer, the interleaving stage, the screen class, the input, and a second number where an exercise reports one (Reckon: on walks back, the distance walked as a share of the true one). For a spreadsheet; cannot be imported.
Research (CSV, in the Lab)	The trial rows plus each item's difficulty and guessing rate under the current item model (chapter 10).

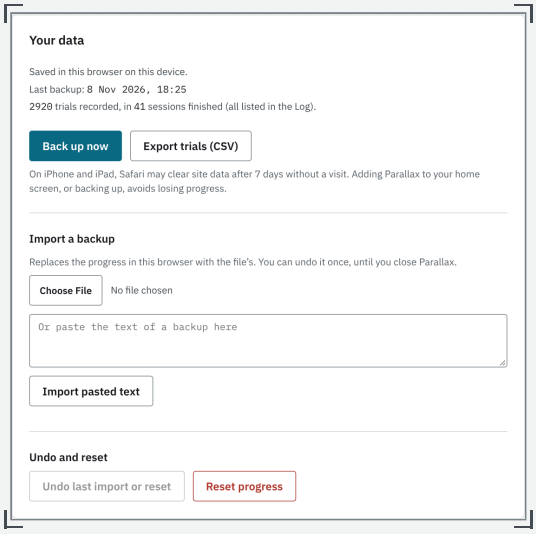


FIG. 9.1 Settings, Your data: back up, export, import, undo and reset.

WHAT A TRIAL ROW DOES NOT HOLD

No name, no account, no device identifier, no location; only the screen class (small or large) and the kind of input. A seed and the settings regenerate the exact item, so a row records what you saw without storing the picture. Timestamps and results are still personal: share exports only where you choose to.

IPHONE, IPAD AND IN-APP BROWSERS

Safari may clear a site's data after seven days of use without a visit, and an app's built-in browser, Discord's included, keeps a copy of its own. Adding Parallax to your home screen (Share, then Add to Home Screen), or backing up regularly, avoids losing progress; to move it, back up in one place and import in the other. A backup is the safe choice on every device.

CHAPTER 10

The Lab

The Lab is for people who want to measure, pool data or test a question on themselves: the Threshold protocols, a research export, a readable account of the engine, and an optional self-experiment.

Threshold protocols

Each protocol keeps one setting on a staircase (chapter 4) and lists its rules and your past runs: Track Standard (the speed you track fully right about half the time: a staircase after Legault, Allard and Faubert, 2013, with the spheres, targets and tracking time of Faubert, 2013), Span length (near 71 % right) and Spin deadline. Stereo displays raised thresholds in adults and children (Plourde et al., 2017), so thresholds from different display modes, Parallax's flat screens included, are not comparable.

Research export

One row per trial with the exercise and its item-model version, the settings, the seed (which regenerates the exact item), the outcome and response time, the chance of a right answer the engine predicted, the interleaving stage, the screen class and the input, a second number where an exercise reports one, and the item's difficulty and guessing rate under the current model. Checks with their one-tap answers, Section runs and the tests you noted travel in the JSON backup. Pooled from many people, these rows can refit the hand-set difficulty steps (chapter 8) and test the defaults this guide calls design bets. Nothing is sent anywhere: you choose what to share and where.

The self-experiment

Does training one group of exercises move the Spatial Check more than training the other? If you join, Parallax draws an order at random, and for six weeks Train sessions use one group at a time:

Group A	Rotate, Spin, Fold, Nets, Slice
Group B	Views, Orient, Span, Hidden, Track, Homing, Reckon
Phases	three weeks each, in the drawn order
Checks	two at the start, one at week 3, one at week 6
Analysis	fixed in advance: for each probe, the change while training A minus the change while training B, with its standard error

Familiarity adds more to the first phase, so one person's contrast leans toward the cluster trained first; the random order cancels this only across many people. One person's result is noisy, so expect "unclear". The caveats are real: there is no blinding, and the second phase starts from a trained state, so carry-over can blur the difference. You can leave at any time; your record keeps everything.

How the engine works

A condensed version of chapter 4 and the developer notes: ratings, levels, item choice, the modes and why there are several, how to read the Spatial Check, and what the simulations show.

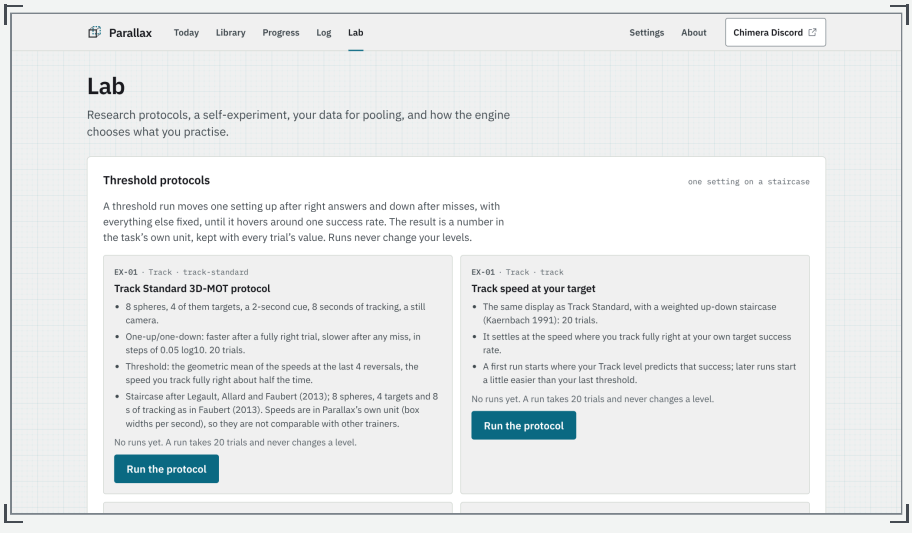


FIG. 10.1 The Lab, opening on the Threshold protocols: their rules, and your runs once there are some.

CHAPTER 11

For contributors

Parallax is a static web app in plain JavaScript: classic scripts on one global, `window.PX`, no build step to run it, no framework, and no network. It has to work from a file, on GitHub Pages, in sandboxed previews and on phones.

Code layout

`index.html`	the app; scripts load in a fixed order
`js/core/`	random numbers, timing, the active-time clock, keys, storage, routing, the shared 3D stage
`js/engine/`	ratings, item choice, modes, the Spatial Check, Section practice, achievements, streaks; pure code that runs in Node
`js/ex/<id>/`	one folder per exercise or probe: gen.js, learn.js, view.js
`js/ui/`	the screens: Today, player, Library, Progress, Log, Lab, Settings, About, share cards, Section practice; overlap.js holds the test-hygiene notes and helper
`css/`	design tokens, app, screens, player and one file per exercise
`tests/node/`	generator, engine and simulation tests
`tests/e2e/`	browser tests (Python Playwright)
`tests/dev/`	the development harness for one exercise
`tools/`	single-file build, screenshots, this guide's PDF, three.js vendoring
`docs/`	this guide, figures, ENGINE.md and EXERCISE-GUIDE.md
`vendor/, fonts/`	three.js (with its licence), IBM Plex (OFL)

House rules

- No claim beyond the evidence: no IQ, no "smarter", no "mastered".
- No clones of clinical subtests and no copied test items.
- Every item verified by a solver; shortcuts near chance.
- Motion in units per second from frame timestamps; the answer clock starts at the painted frame; stuttered trials never count.
- Keyboard everywhere, 44-pixel targets, colour never alone, no flashing faster than 3 Hz; every scored item fits a 360 × 640 screen.
- Test formats named only to describe them, with the non-affiliation line; rules in Parallax's own words.

How an exercise is built

Each exercise is three files plus tests, and `docs/dev/EXERCISE-GUIDE.md` walks through one (Rotate) in full:

- **gen.js** (pure): the definition (code, abilities, difficulty dimensions with justified steps, variants, guessing rate), a generator that never throws, an independent validator that re-derives the key by brute force, a scorer, and a shared model for the view and the tests.
- **learn.js** (data): Watch, Try, Go; labelled parts; tips; two strategy cards and when to switch; the "where it is used" line.
- **view.js**: drawing, options, reveal, show me and demo, built from the same model that grades the answer.
- **css/ex/<id>.css**, **tests/node/<id>.test.mjs** (10,000 seeds or more: the key right, every other answer wrong, determinism, time per item, shortcut solvers including ones that never look at the question) and **tests/e2e/harness_<id>.py**.

Try an exercise alone in the harness: `tests/dev/harness.html?ex=rotate&seed=7&p=angle:4`. ENGINE.md explains every formula and default of the engine.

Ideas welcome

New exercises, better items, translations, accessibility fixes and data analyses are all welcome. Talk it over first on the Chimera Discord, discord.gg/chmr, so work is not duplicated.

Running the tests

One command runs the whole suite: the Node tests first, then every browser test through `pwslot.sh`, two at a time, with a summary in `tests/e2e/out/run_all.json`. `RUN_ALL_SKIP=longrun` leaves out the long bot run.

```
npm install
pip install playwright && python -m playwright install chromium
tests/run_all.sh                                        # everything
```

Node: generators, engine, simulations

```
npm test                                                # every exercise and probe, the engine
node tests/node/run.mjs rotate                          # one exercise
node tests/node/sim.mjs --learners 200                  # simulated learners over 60 days
```

Browser: Python Playwright, headless Chromium

```
tests/e2e/pwslot.sh python3 tests/e2e/shell.py          # app shell, Chimera pop-up, settings
tests/e2e/pwslot.sh python3 tests/e2e/play.py           # the player, feedback, show me, sessions
tests/e2e/pwslot.sh python3 tests/e2e/flows.py          # tour, Spatial Check, Daily Challenge
tests/e2e/pwslot.sh python3 tests/e2e/screens.py        # Library, Progress, Log, Lab, share cards
tests/e2e/pwslot.sh python3 tests/e2e/clock.py          # the clock, in real time (about 30 min)
tests/e2e/pwslot.sh python3 tests/e2e/fit.py            # every scored item fits 360 × 640
tests/e2e/pwslot.sh python3 tests/e2e/section.py        # Section practice end to end
tests/e2e/pwslot.sh python3 tests/e2e/overlap.py        # test hygiene: notes, helper, check question
tests/e2e/pwslot.sh python3 tests/e2e/harness_rotate.py # one exercise in the dev harness (any id)
tests/e2e/pwslot.sh python3 tests/e2e/single.py         # the single-file build
tests/e2e/pwslot.sh python3 tests/e2e/longrun.py        # a bot plays 300 trials in each mode
```

`pwslot.sh` lets at most two headless browsers run at once on one machine; more make every test slow and flaky. Screenshots land in `tests/e2e/out/`: look at them.

Builds

```
node tools/build-single.mjs                             # dist/Parallax.html, one file
tests/e2e/pwslot.sh python3 tools/shots.py              # the figures in docs/shots/
tests/e2e/pwslot.sh python3 tools/build-pdf.py          # this guide, docs/Parallax-Guide.pdf
```

The single file inlines every script, stylesheet and font, so it works offline, from disk and in sandboxed previews. The guide is HTML printed by Chromium; its build fails when any page overflows, and it checks that every link in the PDF is clickable.

Publishing on GitHub Pages

Push the repository to GitHub, then in the repository's **Settings → Pages** choose **Deploy from a branch**, the main branch and the root folder. The site serves `index.html` as it is; nothing to build.

Licences

Code	MIT licence
This guide's text	CC BY-SA 4.0 (LICENSE-GUIDE.md)
three.js	MIT licence, vendor/LICENSE-three.txt
IBM Plex Sans and Mono	SIL Open Font License 1.1, fonts/OFL.txt

CHAPTER 12

Credits and the Chimera community

Parallax was built by the Chimera community on Discord. Come and say hi.

Share your results and your Daily Challenge lines, swap challenge codes with friends, compare strategies, tell us what confused you, and suggest exercises you would like to train. Parallax gets better when people who use it say what works.

Join the Chimera Discord

https://discord.gg/chmr

If a link does not open, type the address into your browser: discord.gg/chmr.

CREDITS

Parallax, its exercises, engine, tests and this guide were made for and with the Chimera community. The research it rests on is listed in chapter 13; thank you to the researchers who publish open data and open materials.

3D rendering: three.js (MIT). Typefaces: IBM Plex Sans and IBM Plex Mono by IBM (SIL OFL 1.1).

THIS GUIDE

Version 1.1.0, 13 November 2026 (first published for 1.0.0, 6 November 2026). Text licensed CC BY-SA 4.0: you may share and adapt it, with credit, under the same licence. Corrections are welcome, especially about the research: if a number here is wrong, tell us on the Discord.

CHAPTER 13

References

Every source cited in this guide. Details that could not be confirmed against a primary source, such as some page numbers, are left out; n.d. marks a year that could not be confirmed.

American Dental Association (2026). *DAT Validity Study*, March 2026 (read through its PDF on ada.org). The ADA's page for the test: ada.org/en/education-careers/dental-admission-test

Andersen, E., O'Rourke, E., Liu, Y.-E., Snider, R., Lowdermilk, J., Truong, D., Cooper, S., & Popović, Z. (2012). The impact of tutorials on games of varying complexity. *Proceedings of CHI '12*, 59–68. doi.org/10.1145/2207676.2207687

Bai, S., Hew, K. F., & Huang, B. (2020). Does gamification improve student learning outcome? Evidence from a meta-analysis and synthesis of qualitative data in educational contexts. *Educational Research Review, 30*, 100322. doi.org/10.1016/j.edurev.2020.100322

Barnett, S. M., & Ceci, S. J. (2002). When and where do we apply what we learn? A taxonomy for far transfer. *Psychological Bulletin, 128*(4), 612–637. doi.org/10.1037/0033-2909.128.4.612

Bediou, B., Adams, D. M., Mayer, R. E., Tipton, E., Green, C. S., & Bavelier, D. (2018). Meta-analysis of action video game impact on perceptual, attentional, and cognitive skills. *Psychological Bulletin, 144*(1), 77–110. doi.org/10.1037/bul0000130

Bediou, B., Rodgers, M. A., Tipton, E., Mayer, R. E., Green, C. S., & Bavelier, D. (2023). Effects of action video game play on cognitive skills: A meta-analysis. *Technology, Mind, and Behavior, 4*(1). doi.org/10.1037/tmb0000102

Bennett-Pierre, G., Shipley, T. F., Newcombe, N. S., & Gunderson, E. A. (2025). Developing a novel measure of non-rigid, ductile spatial skill. *Cognitive Research: Principles and Implications, 10*, 13. doi.org/10.1186/s41235-025-00621-w

Brandt, M. G., & Davies, E. T. (2006). Visual-spatial ability, learning modality and surgical knot tying. *Canadian Journal of Surgery, 49*(6), 412–416.

Brunmair, M., & Richter, T. (2019). Similarity matters: A meta-analysis of interleaved learning and its moderators. *Psychological Bulletin, 145*(11), 1029–1052. doi.org/10.1037/bul0000209

Burte, H., Gardony, A. L., Hutton, A., & Taylor, H. A. (2018). Knowing when to fold 'em: Problem attributes and strategy differences in the Paper Folding test. *Personality and Individual Differences.* doi.org/10.1016/j.paid.2018.08.009

Canivez, G. L. (2010). Review of the WAIS-IV [test review; author's copy]. ux1.eiu.edu/~glcanivez (PDF)

Clement, B., Roy, D., Oudeyer, P.-Y., & Lopes, M. (2015). Multi-armed bandits for intelligent tutoring systems. *Journal of Educational Data Mining*. arxiv.org/abs/1310.3174

Cohen, C. A., & Hegarty, M. (2007). Sources of difficulty in imagining cross sections of 3D objects. *Proceedings of the 29th Annual Conference of the Cognitive Science Society.* escholarship.org/uc/item/98p7t49b

Cohen, C. A., & Hegarty, M. (2014). Visualizing cross sections: Training spatial thinking using interactive animations and virtual objects. *Learning and Individual Differences, 33*, 63–71. doi.org/10.1016/j.lindif.2014.04.002

Croom, S., & Firestone, C. (2024). Tangled physics: Knots strain intuitive physical reasoning. *Open Mind.* pmc.ncbi.nlm.nih.gov/articles/PMC11495958

De Lillo, C., Kirby, M., & Poole, D. (2016). Spatio-temporal structure, path characteristics, and perceptual grouping in immediate serial spatial recall. *Frontiers in Psychology, 7*, 1686. doi.org/10.3389/fpsyg.2016.01686

Deci, E. L., Koestner, R., & Ryan, R. M. (1999). A meta-analytic review of experiments examining the effects of extrinsic rewards on intrinsic motivation. *Psychological Bulletin, 125*(6), 627–668. doi.org/10.1037/0033-2909.125.6.627

Dillon, M. R., Duyck, M., Dehaene, S., Amalric, M., & Izard, V. (2019). Geometric categories in cognition. *Journal of Experimental Psychology: Human Perception and Performance.* doi.org/10.1037/xhp0000663

Doll, R. J., Veltink, P. H., & Buitenweg, J. R. (2015). Observation of time-dependent psychophysical functions and accounting for threshold drifts. *Attention, Perception, & Psychophysics, 77*(4), 1440–1447. doi.org/10.3758/s13414-015-0865-x

Embodied Navigation (2025). *Embodied Navigation: Whole-body movement drives path integration in large-scale free-walking virtual reality* [preprint; authors not recorded]. bioRxiv. biorxiv.org/content/10.1101/2025.11.05.686699

Faubert, J. (2013). Professional athletes have extraordinary skills for rapidly learning complex and neutral dynamic visual scenes. *Scientific Reports, 3*, 1154. doi.org/10.1038/srep01154

Federal Trade Commission (2022). *Health Products Compliance Guidance.* ftc.gov/business-guidance/resources/health-products-compliance-guidance

Fischer, G. H. (1973). The linear logistic test model as an instrument in educational research. *Acta Psychologica, 37.*

Foroughi, C. K., Monfort, S. S., Paczynski, M., McKnight, P. E., & Greenwood, P. M. (2016). Placebo effects in cognitive training. *PNAS, 113*(27), 7470–7474. doi.org/10.1073/pnas.1601243113

Ganis, G., & Kievit, R. (2015). A new set of three-dimensional shapes for investigating mental rotation processes: Validation data and stimulus set. *Journal of Open Psychology Data, 3*, e3. doi.org/10.5334/jopd.ai

Gilligan-Lee, K. A., Hawes, Z. C. K., Williams, A. Y., Farran, E. K., & Mix, K. S. (2023). Hands-on: Investigating the role of physical manipulatives in spatial training. *Child Development.* doi.org/10.1111/cdev.13963

Harris, D. J., Wilson, M. R., Smith, S. J. R., Meder, N., & Vine, S. J. (2020). Testing the effects of 3D multiple object tracking training on near, mid and far transfer. *Frontiers in Psychology, 11*, 196. doi.org/10.3389/fpsyg.2020.00196

Hausknecht, J. P., Halpert, J. A., Di Paolo, N. T., & Moriarty Gerrard, M. O. (2007). Retesting in selection: A meta-analysis of coaching and practice effects for tests of cognitive ability. *Journal of Applied Psychology, 92*(2), 373–385. doi.org/10.1037/0021-9010.92.2.373

DAT® and Dental Admission Test are marks of the American Dental Association. Parallax is not affiliated with or endorsed by the ADA. Every item is generated by the app; none is taken from the DAT.

Hawes, Z. C. K., Gilligan-Lee, K. A., & Mix, K. S. (2022). Effects of spatial training on mathematics performance: A meta-analysis. *Developmental Psychology, 58*(1), 112–137. doi.org/10.1037/dev0001281

He, C., Chrastil, E. R., & Hegarty, M. (2022). A new psychometric task measuring spatial perspective taking in ambulatory virtual reality. *Frontiers in Virtual Reality, 3*, 971502. doi.org/10.3389/frvir.2022.971502

Hegarty, M. (2004). Mechanical reasoning by mental simulation. *Trends in Cognitive Sciences, 8*(6), 280–285. doi.org/10.1016/j.tics.2004.04.001

Hegarty, M. (2018). Ability and sex differences in spatial thinking: What does the mental rotation test really measure? *Psychonomic Bulletin & Review, 25*, 1212–1219. doi.org/10.3758/s13423-017-1347-z

Hegarty, M., & Waller, D. (2004). A dissociation between mental rotation and perspective-taking spatial abilities. *Intelligence, 32*(2), 175–191. doi.org/10.1016/j.intell.2003.12.001

Höffler, T. N., & Leutner, D. (2007). Instructional animation versus static pictures: A meta-analysis. *Learning and Instruction, 17*(6), 722–738. doi.org/10.1016/j.learninstruc.2007.09.013

Information Commissioner's Office (n.d.). *Age appropriate design: a code of practice for online services*, standard 5: Detrimental use of data. ico.org.uk (children's code, standard 5)

Jansen, B. R. J., Louwerse, J., Straatemeier, M., Van der Ven, S. H. G., Klinkenberg, S., & Van der Maas, H. L. J. (2013). The influence of experiencing success in math on math anxiety, perceived math competence, and math performance. *Learning and Individual Differences, 24*, 190–197. doi.org/10.1016/j.lindif.2012.12.014

Kaernbach, C. (1991). Simple adaptive testing with the weighted up-down method. *Perception & Psychophysics, 49*(3), 227–229. doi.org/10.3758/BF03214307

Kalyuga, S., Ayres, P., Chandler, P., & Sweller, J. (2003). The expertise reversal effect. *Educational Psychologist, 38*(1), 23–31. doi.org/10.1207/S15326985EP3801_4

Keehner, M., Hegarty, M., Cohen, C., Khooshabeh, P., & Montello, D. R. (2008). Spatial reasoning with external visualizations: What matters is what you see, not whether you interact. *Cognitive Science, 32*(7), 1099–1132. doi.org/10.1080/03640210801898177

Kivetz, R., Urminsky, O., & Zheng, Y. (2006). The goal-gradient hypothesis resurrected: Purchase acceleration, illusionary goal progress, and customer retention. *Journal of Marketing Research, 43*(1), 39–58. doi.org/10.1509/jmkr.43.1.39

Kluger, A. N., & DeNisi, A. (1996). The effects of feedback interventions on performance: A historical review, a meta-analysis, and a preliminary feedback intervention theory. *Psychological Bulletin, 119*(2), 254–284. doi.org/10.1037/0033-2909.119.2.254

Kozhevnikov, M., & Hegarty, M. (2001). A dissociation between object manipulation spatial ability and spatial orientation ability. *Memory & Cognition, 29*(5), 745–756.

Kurtuluş, A., & Yolcu, B. (2013). A study on sixth-grade Turkish students' spatial visualization ability. *The Mathematics Educator, 22*(2), 82–117.

Lampit, A., Hallock, H., & Valenzuela, M. (2014). Computerized cognitive training in cognitively healthy older adults: A systematic review and meta-analysis of effect modifiers. *PLOS Medicine, 11*(11), e1001756. doi.org/10.1371/journal.pmed.1001756

Legault, I., Allard, R., & Faubert, J. (2013). Healthy older observers show equivalent perceptual-cognitive training benefits to young adults for multiple object tracking. *Frontiers in Psychology, 4*, 323. doi.org/10.3389/fpsyg.2013.00323

Levacher, J., Koch, M., Hissbach, J., Spinath, F. M., & Becker, N. (2022). You can play the game without knowing the rules, but you're better off knowing them. *European Journal of Psychological Assessment.* doi.org/10.1027/1015-5759/a000637

Levitt, H. (1971). Transformed up-down methods in psychoacoustics. *Journal of the Acoustical Society of America, 49*(2B), 467–477. doi.org/10.1121/1.1912375

Liesefeld, H. R., & Janczyk, M. (2019). Combining speed and accuracy to control for speed-accuracy trade-offs(?). *Behavior Research Methods, 51*, 40–60. doi.org/10.3758/s13428-018-1076-x

Lövdén, M., Schaefer, S., Noack, H., Bodammer, N. C., Kühn, S., Heinze, H.-J., Düzel, E., Bäckman, L., & Lindenberger, U. (2012). Spatial navigation training protects the hippocampus against age-related changes during early and late adulthood. *Neurobiology of Aging, 33*(3), 620.e9–620.e22. pub.dzne.de (PDF)

McGrew, K. S. (2017). *CHC broad and narrow ability definitions*, abstracted from Schneider & McGrew (2018). Institute for Applied Psychometrics. iapsych.com/chcdefs.pdf

McLaren, R., Chaudhary, S., Rashid, U., Ravindran, S., & Taylor, D. (2022). Reliability of the triangle completion test in the real-world and in virtual reality. *Frontiers in Human Neuroscience, 16*, 945953. doi.org/10.3389/fnhum.2022.945953

Melby-Lervåg, M., Redick, T. S., & Hulme, C. (2016). Working memory training does not improve performance on measures of intelligence or other measures of "far transfer". *Perspectives on Psychological Science, 11*(4), 512–534. ora.ox.ac.uk (open copy)

Meneghetti, C., Borella, E., & Pazzaglia, F. (2016). Mental rotation training: Transfer and maintenance effects on spatial abilities. *Psychological Research, 80*(1). doi.org/10.1007/s00426-014-0644-7

Meyerhoff, H. S., Papenmeier, F., & Huff, M. (2017). Studying visual attention using the multiple object tracking paradigm: A tutorial review. *Attention, Perception, & Psychophysics, 79*(5), 1255–1274. doi.org/10.3758/s13414-017-1338-1

Miller, D. I., & Halpern, D. F. (2013). Can spatial training improve long-term outcomes for gifted STEM undergraduates? *Learning and Individual Differences, 26*, 141–152. doi.org/10.1016/j.lindif.2012.03.012

Mix, K. S., Hambrick, D. Z., Satyam, V. R., Burgoyne, A. P., & Levine, S. C. (2018). The latent structure of spatial skill: A test of the 2 × 2 typology. *Cognition, 180*, 268–278. doi.org/10.1016/j.cognition.2018.07.012

Molloy, K., Moore, D. R., Sohoglu, E., & Amitay, S. (2012). Less is more: Latent learning is maximized by shorter training sessions in auditory perceptual learning. *PLOS ONE, 7*(5), e36929. doi.org/10.1371/journal.pone.0036929

Nazareth, A., Killick, R., Dick, A. S., & Pruden, S. M. (n.d.). Strategy selection versus flexibility: Eye-trackers and mental rotation. *Journal of Experimental Psychology: Learning, Memory, and Cognition.* Author manuscript: eprints.lancs.ac.uk/89803

Newcombe, N. S., & Shipley, T. F. (2015). Thinking about spatial thinking: New typology, new assessments. In J. S. Gero (Ed.), *Studying Visual and Spatial Reasoning for Design Creativity* (pp. 179–192). Springer. doi.org/10.1007/978-94-017-9297-4_10

Noel, J.-P., Lakshminarasimhan, K. J., Park, H., & Angelaki, D. E. (2020). Increased variability but intact integration during visual navigation in Autism Spectrum Disorder. *PNAS, 117*(20), 11158–11166. doi.org/10.1073/pnas.2000216117

Pelánek, R. (2016). Applications of the Elo rating system in adaptive educational systems. *Computers & Education, 98.* doi.org/10.1016/j.compedu.2016.03.017

Pelánek, R., & Řihák, J. (2018). Analysis and design of mastery learning criteria. *New Review of Hypermedia and Multimedia.* fi.muni.cz/~xpelanek (PDF)

Pelánek, R., Papoušek, J., Řihák, J., Stanislav, V., & Nižnan, J. (2017). Elo-based learner modeling for the adaptive practice of facts. *User Modeling and User-Adapted Interaction.* fi.muni.cz/~xpelanek (PDF)

Perkins, D. N., & Salomon, G. (1988). Teaching for transfer. *Educational Leadership, 46*(1), 22–32. files.ascd.org (PDF)

Plourde, M., Corbeil, M.-E., & Faubert, J. (2017). Effect of age and stereopsis on a multiple-object tracking task. *PLOS ONE, 12*(12), e0188373. doi.org/10.1371/journal.pone.0188373

Porter, J. M., & Magill, R. A. (2010). Systematically increasing contextual interference is beneficial for learning sport skills. *Journal of Sports Sciences.* doi.org/10.1080/02640414.2010.502946

Rajeb, M., Krist, A. T., Shi, Q., Oyeniran, D. O., Wind, S. A., & Lakin, J. M. (2025). Mental rotation performance: Contribution of item features to difficulties and functional adaptation. *Journal of Intelligence, 13*(1), 2. doi.org/10.3390/jintelligence13010002

Raviv, L., Lupyan, G., & Green, S. C. (2022). How variability shapes learning and generalization. *Trends in Cognitive Sciences, 26*(6), 462–483. doi.org/10.1016/j.tics.2022.03.007

Riecke, B. E., & Wiener, J. M. (2007). Point-to-origin experiments in VR revealed novel qualitative errors in visual path integration. Max Planck Institute for Biological Cybernetics (short conference paper). core.ac.uk (PDF)

Rohrer, D., & Taylor, K. (2007). The shuffling of mathematics problems improves learning. *Instructional Science, 35*, 481–498. doi.org/10.1007/s11251-007-9015-8

Romeas, T., Goujat, M., Faubert, J., & Labbé, D. (2025). No transfer of 3D-Multiple Object Tracking training on game performance in soccer: A follow-up study. *Psychology of Sport and Exercise, 76*, 102770. doi.org/10.1016/j.psychsport.2024.102770

Ryan, R. M., Rigby, C. S., & Przybylski, A. (2006). The motivational pull of video games: A self-determination theory approach. *Motivation and Emotion, 30*(4), 344–360. doi.org/10.1007/s11031-006-9051-8

Sailer, M., & Homner, L. (2020). The gamification of learning: A meta-analysis. *Educational Psychology Review, 32*(1), 77–112. doi.org/10.1007/s10648-019-09498-w

Sailer, M., Hense, J. U., Mayr, S. K., & Mandl, H. (2017). How gamification motivates: An experimental study of the effects of specific game design elements on psychological need satisfaction. *Computers in Human Behavior, 69*, 371–380. doi.org/10.1016/j.chb.2016.12.033

Sala, G., & Gobet, F. (2019). Cognitive training does not enhance general cognition. *Trends in Cognitive Sciences, 23*(1), 9–20. livrepository.liverpool.ac.uk/3030313

Sala, G., Tatlidil, K. S., & Gobet, F. (2018). Video game training does not enhance cognitive ability: A comprehensive meta-analytic investigation. *Psychological Bulletin, 144*(2), 111–139. doi.org/10.1037/bul0000139

Scharfen, J., Peters, J. M., & Holling, H. (2018). Retest effects in cognitive ability tests: A meta-analysis. *Intelligence, 67*, 44–66. doi.org/10.1016/j.intell.2018.01.003

Schneider, B., Becker, N., Krieger, F., Spinath, F. M., & Sparfeldt, J. R. (2020). Teaching the underlying rules of figural matrices in a short video increases test scores. *Intelligence, 82*, 101473. doi.org/10.1016/j.intell.2020.101473

Schneider, W. J., & McGrew, K. S. (2018). The Cattell-Horn-Carroll theory of cognitive abilities. In D. P. Flanagan & E. M. McDonough (Eds.), *Contemporary Intellectual Assessment* (4th ed.). Guilford.

Shepard, R. N., & Feng, C. (1972). A chronometric study of mental paper folding. *Cognitive Psychology, 3*, 228–243.

Silverman, J., & Barasch, A. (2023). On or off track: How (broken) streaks affect consumer decisions. *Journal of Consumer Research, 49*(6), 1095–1117. doi.org/10.1093/jcr/ucac029

Simons, D. J., Boot, W. R., Charness, N., Gathercole, S. E., Chabris, C. F., Hambrick, D. Z., & Stine-Morrow, E. A. L. (2016). Do "brain-training" programs work? *Psychological Science in the Public Interest, 17*(3), 103–186. doi.org/10.1177/1529100616661983

Sommer, M., & Arendasy, M. (2025). Automatic- and transformer-based automatic item generation: A critical review. *Journal of Intelligence, 13*(8), 102. mdpi.com/2079-3200/13/8/102

Sorby, S. A. (2009). Educational research in developing 3-D spatial skills for engineering students. *International Journal of Science Education, 31*(3), 459–480. doi.org/10.1080/09500690802595839

Sorby, S. A., & Veurink, N. (n.d.). Long-term results from spatial skills intervention. ASEE Engineering Design Graphics Division paper. edgd.asee.org (PDF)

Stieff, M., & Uttal, D. (2015). How much can spatial training improve STEM achievement? *Educational Psychology Review, 27*, 607–615. doi.org/10.1007/s10648-015-9304-8

Stieff, M., Dixon, B. L., Ryu, M., Kumi, B. C., & Hegarty, M. (2014). Strategy training eliminates sex differences in spatial problem solving in a STEM domain. *Journal of Educational Psychology, 106*(2), 390–402. doi.org/10.1037/a0034823

te Nijenhuis, J., van Vianen, A. E. M., & van der Flier, H. (2007). Score gains on g-loaded tests: No g. *Intelligence, 35.* doi.org/10.1016/j.intell.2006.07.006

Treviño, M., Zhu, X., Lu, Y. Y., Scheuer, L. S., Passell, E., Huang, G. C., Germine, L. T., & Horowitz, T. S. (2021). How do we measure attention? Using factor analysis to establish construct validity of neuropsychological tests. *Cognitive Research: Principles and Implications, 6*, 51. doi.org/10.1186/s41235-021-00313-1

Tuzun, R. E., & Sikora, A. S. (n.d.). Verification of the Jones unknot conjecture up to 24 crossings. arXiv:2003.06724. arxiv.org/abs/2003.06724

Uttal, D. H., Meadow, N. G., Tipton, E., Hand, L. L., Alden, A. R., Warren, C., & Newcombe, N. S. (2013). The malleability of spatial skills: A meta-analysis of training studies. *Psychological Bulletin, 139*(2), 352–402. doi.org/10.1037/a0028446

Vater, C., Gray, R., & Holcombe, A. O. (2021). A critical systematic review of the Neurotracker perceptual-cognitive training tool. *Psychonomic Bulletin & Review, 28*(5), 1458–1483. doi.org/10.3758/s13423-021-01892-2

Voyer, D. (2011). Time limits and gender differences on paper-and-pencil tests of mental rotation: A meta-analysis. *Psychonomic Bulletin & Review, 18*(2), 267–277. doi.org/10.3758/s13423-010-0042-0

Wilson, R. C., Shenhav, A., Straccia, M., & Cohen, J. D. (2019). The Eighty Five Percent Rule for optimal learning. *Nature Communications, 10*, 4646. doi.org/10.1038/s41467-019-12552-4

Wisniewski, B., Zierer, K., & Hattie, J. (2020). The power of feedback revisited: A meta-analysis of educational feedback research. *Frontiers in Psychology, 10*, 3087. doi.org/10.3389/fpsyg.2019.03087

Wright, R., Thompson, W. L., Ganis, G., Newcombe, N. S., & Kosslyn, S. M. (2008). Training generalized spatial skills. *Psychonomic Bulletin & Review, 15*(4), 763–771. doi.org/10.3758/PBR.15.4.763

Yang, W., Liu, H., Chen, N., Xu, P., & Lin, X. (2020). Is early spatial skills training effective? A meta-analysis. *Frontiers in Psychology, 11*, 1938. doi.org/10.3389/fpsyg.2020.01938

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